Morality and Sentiment Analysis of Twitter Data based on Indian Context using Deep Learning

A
Project Report
Submitted in partial fulfilment of the
Requirements for the award of the Degree of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

By

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DECLARATION BY THE CANDIDATE

We, **Anurag Sai Y, Ashish Sadhu** and **Madhav NLV** bearing hall ticket number, **1602-21-737-009, 1602-21-737-011** and **1602-21-737-031** hereby declare that the project report entitled **Morality and Sentiment Analysis of Twitter Data based on Indian Context using Deep Learning** under the guidance of **R DHARMA REDDY, Associate professor**, Department of Information Technology, Vasavi College of Engineering, Hyderabad, is submitted in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Information Technology**

This is a record of bonafide work carried out by us and the results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

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This is to certify that the project entitled **Morality and Sentiment Analysis of Twitter Data based on Indian Context using Deep Learning** being submitted by **Anurag Sai Y, Ashish Sadhu** and **Madhav NLV** bearing **1602-21-737-009, 1602-21-737-011** and **1602-21-737-031** in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Information Technology is a record of bonafide work carried out by them under my guidance.

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ACKNOWLEDGEMENT

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ABSTRACT

In the contemporary era, social media has emerged as an essential medium for individuals to share their views on sociopolitical matters. This research suggests a morality and sentiment-based stance analysis framework specific to Twitter discourse in India, where such matters are culturally nuanced and insufficiently investigated in computational research. We present a new dataset of 6,011 tweets on prominent Indian issues, such as the Citizenship Amendment Act (CAA), Uniform Civil Code (UCC), LGBTQ rights, Brain Drain, Electronic Voting Machines (EVM), Animal Sacrifice, and Anti-Conversion Laws. These issues have been given limited attention in terms of moral foundation theory and sentiment analysis.

While the majority of the recent models have been tested on global datasets, including Twitter datasets, there is not much work done comparing different models using Twitter data specific to the Indian context. This work attempts to fill this gap by focusing on data that captures the sociopolitical climate of India, thus offering a more realistic and culture-sensitive assessment.

To evaluate how effectively deep learning models can perceive sentiment and moral cues, we used and compared three models: Bidirectional LSTM (Bi-LSTM) with TF-IDF embeddings, Bidirectional GRU (Bi-GRU) with GloVe embeddings, and transformer-based BERT. A k-fold cross-validation strategy was used to guarantee good performance on tasks.

The findings emphasize BERT's superiority in sentiment and morality classification with 99.19% and 91.73% accuracies, respectively. Bi-LSTM models recorded mid-range scores of 78.24% and 62.44% for sentiment and morality, respectively. Bi-GRU model with GloVE encodings, despite being light and efficient, recorded about 71.93% and 52.67% respectively. GRU model with TF-IDF encoding have the reading below the Bi-GRU model with 70.01% and 45.73% respectively. Finally, the implantation of LSTM with GloVe embeddings proves with the least readings of all the working models with accuracy of 32.61% and 28.34% respectively. The findings emphasize the strength of contextual embeddings in BERT while recognizing the strength of recurrent networks in culturally diverse, low-resource settings such as India.

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- **BERT** – Bidirectional Encoder Representations from Transformers
- **Bi-LSTM -** Bidirectional Long Short-Term Memory Network
- **Bi-GRU –** Bi-Directional Gated Recurrent Unit
- **CAA -** Citizenship Amendment Act
- **UCC-** Uniform Civil Code
- **LGBTQ -** lesbian, gay, bisexual, transgender and queer
- **MFT –** Moral Foundation Theory

1. INTRODUCTION

1.1. Problem Statement

In the era of pervasive digital communication, social media platforms like Twitter have become critical spaces for expressing public sentiment and moral viewpoints on pressing sociopolitical issues. While sentiment analysis has gained significant attention in natural language processing, the intersection of morality and stance detection in the Indian socio-cultural context remains underexplored. Traditional sentiment models often fail to capture the underlying moral dimensions of user opinions, which are essential for a deeper understanding of public discourse.

Furthermore, there is a lack of domain-specific annotated datasets and empirical evaluations of deep learning models for morality-based classification in Indian scenarios. This creates a need for a comprehensive, contextual, and accurate user profiling framework that can effectively analyse both sentiment and moral alignment from real-world social media data.

To address this gap, this work aims to:

1. Develop a custom morality-annotated tweet corpus based on Indian socio-political topics.
2. Evaluate the performance of deep learning architectures, specifically Bi-LSTM and BERT, on sentiment and moral foundation classification.
3. Benchmark model accuracy and generalizability using k-fold cross-validation to ensure robustness.

1.2. Motivation

With the exponential growth of social media usage, platforms like Twitter have evolved into influential arenas for sociopolitical debate and public discourse. In a diverse and complex democracy like India, opinions shared online often reflect not just emotional reactions, but deeper moral and ideological stances rooted in cultural, religious, and political contexts. Understanding these perspectives is vital for a range of applications—from policymaking and media analysis to social research and digital governance.

While sentiment analysis has become a mainstream task in natural language processing, it provides only a surface-level understanding of user intent. The moral reasoning that drives support or opposition toward key issues often goes unnoticed in traditional approaches. This gap limits the interpretability and effectiveness of models used for opinion mining in sensitive, multifaceted domains such as citizenship laws, gender rights, or religious policies.

Furthermore, existing datasets and models are predominantly trained on Western-centric content, making them less applicable to Indian discourse, where language usage, idiomatic expressions, and moral framing differ substantially. There is a pressing need to build context-aware, culturally grounded models capable of decoding both sentiment and moral undertones in user-generated content.

This project is driven by the need to:

- Develop a tailored dataset that reflects Indian sociopolitical discourse.
- Build models that go beyond polarity classification to capture moral depth.
- Empower researchers and analysts with tools that can interpret ideological trends in digital conversations with high accuracy.

By combining sentiment analysis with morality classification using modern deep learning architectures like BERT and Bi-LSTM, this work seeks to bridge the gap between computational linguistics and ethical perspective modelling in the Indian context.

1.3. Scope & Objectives of the Proposed Work

Scope of the Proposed Work

This research focuses on the development of a deep learning-based framework for sentiment and morality classification from social media text, specifically Twitter posts related to Indian sociopolitical discourse. The scope encompasses:

- Designing a domain-specific tweet corpus annotated for sentiment polarity and moral foundations.
- Analysing and interpreting the accuracy, precision, recall, and F1-scores of both models on sentiment and moral classification tasks.
- Highlighting the applicability of contextual language models in understanding complex public discourse in culturally rich settings.

This work is restricted to text-based analysis of English-language tweets and focuses on Indian themes, while not extending to multilingual or multimodal inputs.

Objectives of the Proposed Work:

1. To curate a labelled dataset of 6,011 Indian tweets, annotated for sentiment (positive, negative, neutral) and morality (based on Moral Foundations Theory).
2. To implement Bi-LSTM models using TF-IDF embeddings and evaluate their performance on sentiment and morality classification tasks.
3. To fine-tune and apply BERT, leveraging its contextual embeddings to improve classification accuracy and extract deeper semantic features.
4. To implement Bi-Directional GRU models integrated with GloVe word embeddings to capture sequential dependencies and semantic relationships in informal Twitter language.
5. To perform k-fold cross-validation for all models to ensure statistical validity and generalizability of results.
6. To compare and analyze the performance of Bi-LSTM, Bi-GRU, and BERT models across evaluation metrics such as accuracy, F1-score, precision, and recall.
7. To demonstrate the superiority of transformer-based models (BERT) in capturing nuanced moral and emotional stances in culturally contextualized digital conversations, while also evaluating the effectiveness of recurrent models like Bi-GRU for resource-constrained environments.

1.4. Organization of the Report

1. Introduction
Problem Statement

Despite the growing use of sentiment analysis on social media content, existing models often fail to capture the moral and ideological dimensions underlying user opinions, especially within the diverse socio-political context of India. There is a lack of domain-specific datasets and comprehensive models that integrate both sentiment and morality and stance classification. This limits the ability to perform accurate and insightful user profiling. Therefore, this work aims to develop a robust framework that combines sentiment and moral foundation analysis using deep learning models on a curated corpus of Indian tweets.

Motivation
Social media platforms like Twitter are rich sources of public opinion, often reflecting not just sentiments but also deep moral and ideological stances. However, existing sentiment analysis models overlook these moral cues, especially in culturally diverse contexts like India. This motivates the need for context-aware models and datasets that can accurately analyze both sentiment and morality, enabling more meaningful and insightful user profiling

Scope & Objectives

This work aims to develop a deep learning-based framework for sentiment and moral foundation analysis using Twitter data focused on Indian sociopolitical topics. A custom corpus of 6,011 tweets was created and annotated for sentiment and morality.

The project evaluates and compares Bi-LSTM with TF-IDF embeddings, Bi-Directional GRU with GloVe embeddings, and BERT models, using k-fold cross-validation to ensure robustness and generalizability across sentiment and morality classification tasks.

2. Literature Survey

Primarily the survey focused on similar work [1] where majorly focused on morality classification and taken as base work. Explored Hindi based Indian dataset [3]. Researching on various models for sentiment analysis to work-on such as Moral-BERT [4] and Bi-LSTM [9] and understanding the working of models to adapt to our built Indian-contextual dataset. Also understanding the working of Stance classification Application [5]. Finally explored over the similar framework which combines sentiment and morality-based analysis but over a Persian dataset [12].

3. Proposed Model

Methodology

The proposed framework employs a combination of moral foundation theory and advanced stance classification algorithms. By analyzing user-generated content through these lenses, the system identifies and manages morally sensitive discussions. The model incorporates data from diverse social media platforms for comprehensive coverage.

Hardware/Software Specifications

The framework utilizes:

- Machine learning libraries (e.g. PyTorch) for building classification models.
- Natural language processing (NLP) tools (e.g., spaCy, Hugging Face) for text analysis.

4. Experimental Results and Analysis

A dataset comprising user-generated content on sensitive topics was processed to train and validate the model. Results demonstrate high accuracy in identifying morally charged content and differentiating between divisive and constructive discussions. Case studies highlight the system's effectiveness in promoting healthier digital communication.

5. Conclusion & Future Scope

The stance classification framework for Twitter uses advanced models to detect support, opposition, or neutrality in user content, addressing challenges like informal language and understanding the ideology of the user by moral foundation classification. Future improvements include multilingual support, real-time processing, and integration for dynamic content moderation over different social media platforms (majorly Twitter) preventing a building of toxic discourse

2. LITERATURE SURVEY

S.no	Title	Author	Summary & Scope
1	**Morality Classification in Natural Language Text**	Matheus Camasmie Pavan, Vitor Garcia dos Santos, Alex Gwo Jen Lan	Explored morality classification in text using machine learning, applying moral foundation theory to identify values like care, fairness, and loyalty. Their approach, using transformers and fine-tuned models, showed strong accuracy, though it focused on Portuguese datasets, limiting applicability to other cultural contexts.
2	**Hostility Detection Dataset in Hindi**	Mohit Bhardwaj, Md Shad Akhtar, Asif Ekbal, Amitava Das, Tanmoy Chakraborty	developed a multi-BERT model for detecting hostility in Hindi text, effectively identifying hate speech, fake news, and offensive language. But in this case, there was no extensive use of Twitter data.
3	**Moral BERT: A Fine-Tuned Language Model for Capturing Moral Values in Social Discussions**	Gayathri Rajendran, Bhadrachalam Chitturi, Prabaharan Poornachandran	This approach, which utilizes a fine-tuned BERT model, effectively identifies and classifies moral stances such as fairness, care, and loyalty within online content. We aimed to enhance our understanding of how moral framing influences online discourse.
4	**Addressing Event-Driven Concept Drift in Twitter Stream: A Stance Detection Application:**	Bechini, Alessio	This framework is particularly valuable for real-time classification tasks, improving the accuracy of stance detection on platforms like Twitter. We worked on improving the accuracy for the models presented in this paper.

5	**Moral Foundations Twitter Corpus: A Collection of 35k Tweets Annotated for Moral Sentiment**	Hoover, Joe	The Moral Foundations Twitter Corpus is a dataset of over 35,000 tweets labeled for 10 types of moral sentiment. Each tweet is annotated by at least three trained individuals. Unlike this dataset which is based on contemporary global issues we introduced Indian specific corpus.
6	**An Improved Bi-LSTM Approach for User Stance Detection Based on External Commonsense Knowledge and Environment Information**	Peng Jia, Yajun Du, Jingrong Hu, Hui Li, Xianyong Li and Xiaoliang Chen	We have taken the Study of Bi-LSTM for Stance detection over social networks data.
7	**A transformer-based deep learning model for Persian moral sentiment analysis**	Behnam Karami, Fatemeh Bakouie, and Shahriar Gharibzadeh	This study presents a dataset of 8,000 Persian tweets annotated for moral sentiment using Moral Foundations Theory. It shows that fine-tuning a Persian BERT model gives the best results, with a smaller transformer model also performing well. We are considering a similar framework to apply a 6011 Indian tweet datset.

3. PROPOSED SYSTEM

3.1. Methodology

3.1.1. Block Diagram

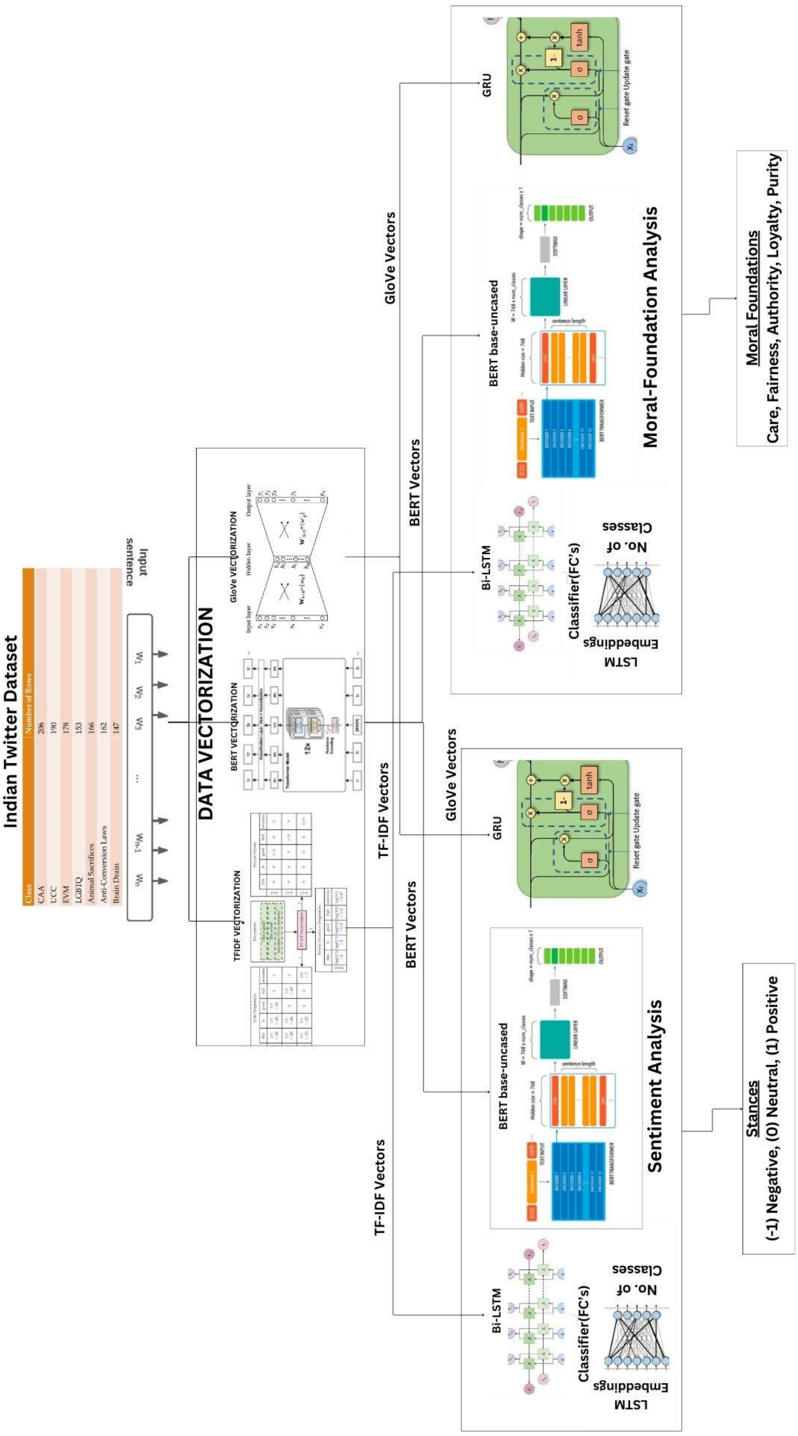

Fig 3.1.1 Block Diagram (Workflow)

Data Collection:

A custom Indian Twitter Dataset is curated, containing 6,011 tweets related to various sociopolitical topics like CAA, Brain Drain, LGBTQ, UCC, etc.

Data Preprocessing & Vectorization:

The raw tweets are cleaned and then transformed using three vectorization techniques:

- TF-IDF Vectorization
- BERT-based Embedding Generation
- GloVe Vectorization

Parallel Pipeline for Analysis:

- **Models Used:**
    - Bi-LSTM model trained on TF-IDF vectors
    - BERT-based classifier trained on BERT embeddings
    - Bi-directional GRU trained on GloVe vectors

The vectorized data is used in two parallel pipelines:

- **Sentiment Analysis Pipeline**
- Classification Output:
    - Sentiment stance:
        - For (+1)
        - Neutral (0)
        - Against (-1)
- **Moral Foundation Analysis Pipeline**
- Classification Output**:**
    - Moral foundations based on **Moral Foundations Theory (MFT)**:
        - Care
        - Fairness
        - Purity
        - Authority
        - Loyalty

Model Evaluation:

- BERT, Bi-GRU and Bi-LSTM models are evaluated using **k-fold cross-validation**.
- Metrics include **accuracy, precision, recall, and F1-score**.

- Results show BERT models outperform both Bi-LSTM, Bi-GRU especially in capturing contextual nuances

3.2. Algorithms

- **BERT Development:**

Create and fine-tune a range of language representation models (BERT) specifically designed to capture and analyse moral sentiment in social discourse.

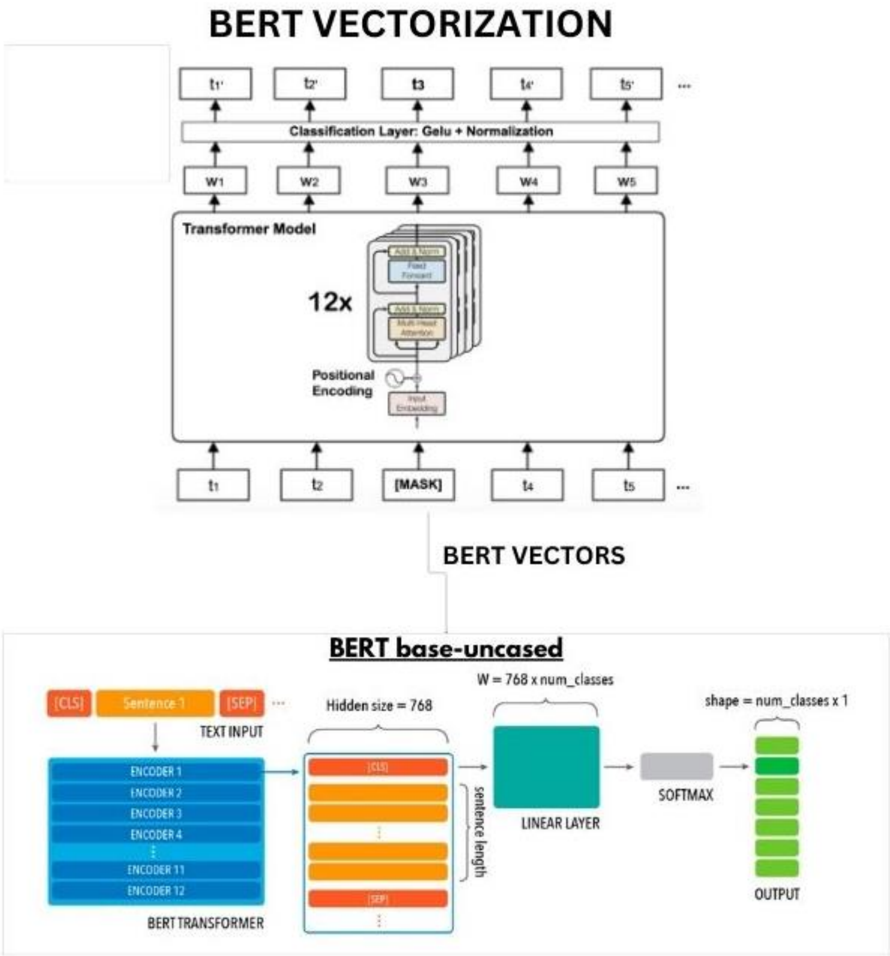

Fig 3.2.1. BERT Model

- **LSTM with Fully Connected Layers (FCs)**:

One of the efficient and better way for feature extraction and sentiment analysis. For given text data extracts temporal dependencies and FCs produce predictions.

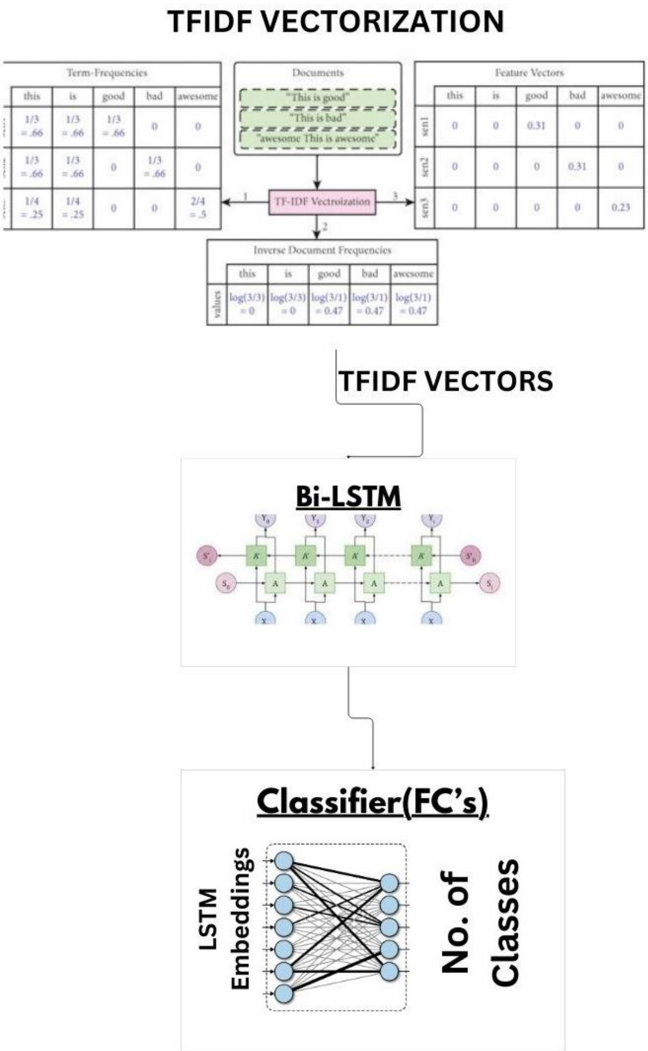

Fig 3.2.2. Bi-LSTM Model

- **Bi-Directional GRU with GloVe embeddings**: Implemented a Bi-GRU model with GloVe embeddings to perform sentiment and moral foundation analysis on Indian Twitter data, effectively capturing contextual and moral cues in socio-political discourse.

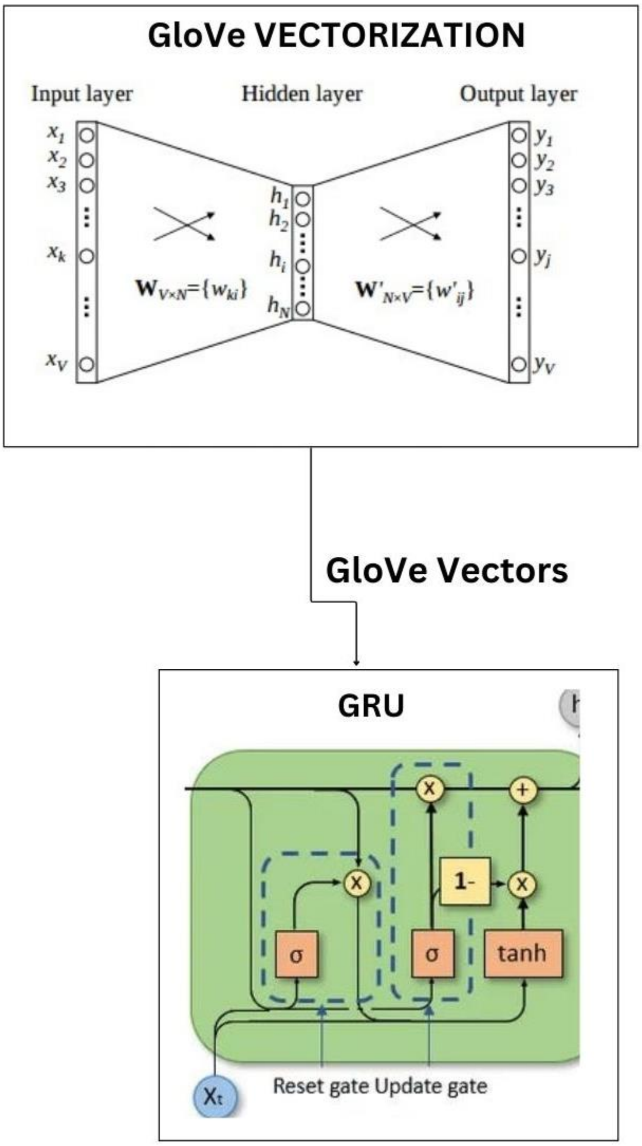

Fig 3.2.3 Bi-GRU Model

- **GRU with TF-IDF Embeddings**: Implemented a GRU model with TF-IDF embeddings to perform sentiment and moral foundation analysis on Indian Twitter data, leveraging statistical word importance to guide the GRU in capturing temporal patterns and thematic structures within socio-political discourse.

4.EXPERIMENTAL SETUP AND RESULTS

4.1. System Specifications

4.1.1. Hardware Specifications:

- **Processor (CPU)**: Intel Core i7
- **Graphics Card (GPU)**: NVIDIA RTX 3060 or higher with 12GB VRAM for deep learning tasks
- **RAM**: Minimum 16GB (32GB recommended for large datasets)

4.1.2. Software Specifications:

- **Operating System**: Windows 11
- **Programming Language**: Python 3.x
- **Libraries**: Pandas, Matplotlib, Seaborn, Sklearn, torch, Transformers, Bert Tokenizer, Bert Model, NumPy

4.2. Description

1) Corpus:

The dataset comprises tweets from Indian social media users, covering diverse sociocultural, political, and economic topics. It includes tweets with hashtags such as #EVM, #CAA, #LGBTQ, #AnimalSacrifice, #UCC, and #BrainDrain, reflecting varied perspectives and discussions within the Indian context.

```python
plt.figure(figsize=(8, 6))
sns.countplot(x='Corpus', data=data)
plt.title('Distribution of Classes', fontsize=16)
plt.xlabel('Corpus', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.tight_layout()
plt.show()
```
[6] ✓ 0.3s

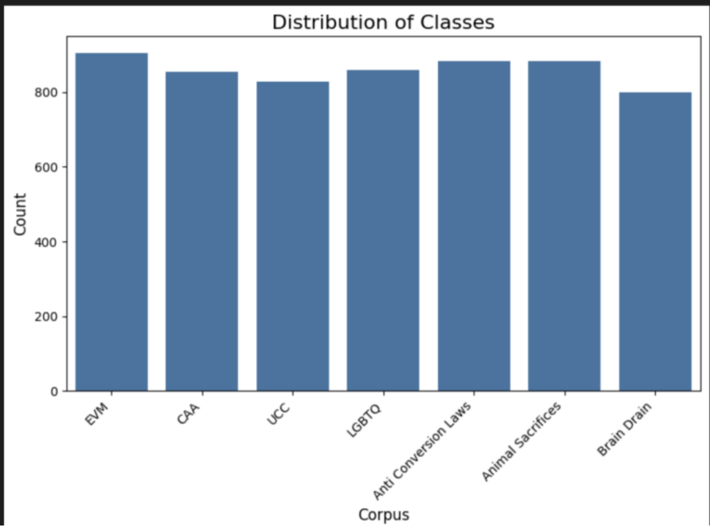

Fig 4.2.1 Distribution of Classes

The above bar chart titled visualizes the count of samples across different topics (or corpora) in the dataset. It shows a relatively balanced distribution of tweets or data points for the following topics: **EVM, CAA, UCC, LGBTQ, Anti Conversion Laws, Animal Sacrifices, Brain Drain**

Each category has approximately 800 to 950 samples, indicating a well-balanced dataset suitable for training classification models without major class imbalance issues.

```python
class_counts = data.iloc[:, 0].value_counts()

print("Number of rows for each class:")
print(class_counts)

total_rows = data.shape[0]
print("\nTotal number of rows:", total_rows)
```
[5] ✓ 0.0s

```
Number of rows for each class:
Corpus
EVM                    904
Animal Sacrifices      882
Anti Conversion Laws   882
LGBTQ                  860
CAA                    855
UCC                    828
Brain Drain            800
Name: count, dtype: int64

Total number of rows: 6011
```

Fig 4.2.2 No. of Tweets

2) Tweets:

The dataset used in this project consists of **6,011 tweets** that are categorized into different social and political issue-based classes. Each tweet corresponds to a specific topic, reflecting public discourse and opinions on that subject. The distribution of tweets across the classes is relatively balanced, ensuring fair representation for training and evaluation.

Class	**Number of Tweets**
EVM	984
Animal Sacrifices	882
Anti Conversion Laws	882
LGBTQ	860
CAA	855
UCC	828
Brain Drain	800
Total	**6,011**

3) Stance:

Categorizes each tweet into one of the three stance labels:

- **Favor (+1)**: Supporting a specific stance target.
- **Against (-1)**: Opposing a specific stance target.
- **Neutral (0)**: Expressing no clear stance.

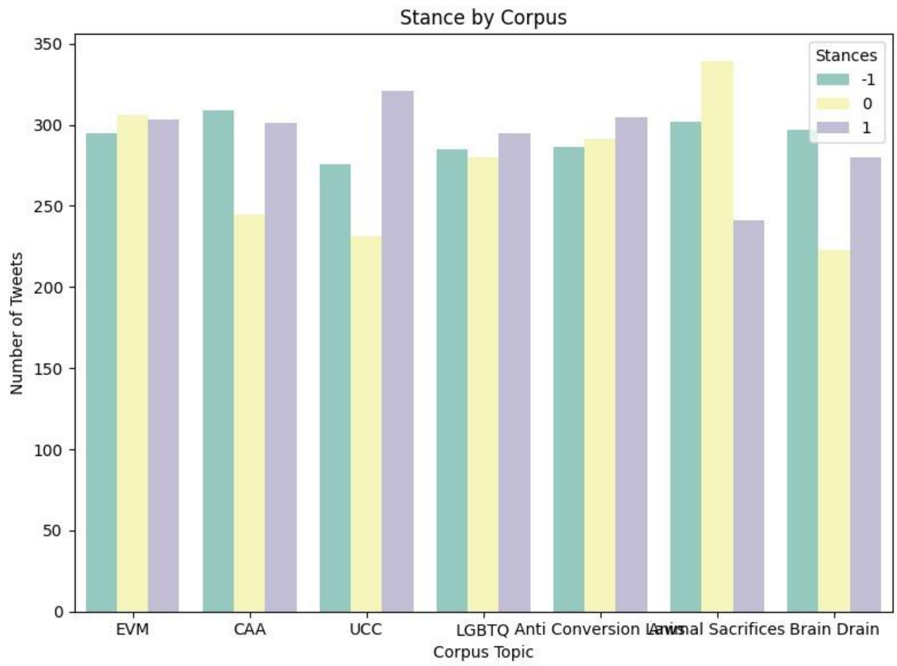

Fig 4.2.3. Stance Distribution by Corpus Topic

The above bar chart shows the distribution of stances on different socio-political issues in the data set. Each issue comprises tweets that fall into one of three stance categories: -1 (Opposing), 0 (Neutral), and 1 (Supporting). This chart gives a clear picture of the different articulations of public opinion across these issues.

Topics such as UCC (Uniform Civil Code) and Brain Drain exhibit a relatively higher proportion of positive (label 1) tweets, indicating a lean towards supporting change or acknowledgment of concern, respectively. In contrast, topics such as Animal Sacrifices and CAA (Citizenship Amendment Act) possess more skewed distributions, with a strong lean towards neutral (0) and negative (-1) stances, perhaps because of the controversial and sensitive nature of debate on these topics.

The issues of Anti-Conversion Laws and LGBTQ rights have a somewhat evenly divided set of positions on the three options. EVMs also experience a near equally divided set of positions among the three stances.

4) Foundation of Tweet:

Identifies the moral or emotional foundation driving the tweet:

- **Care**: Emphasis on empathy or concern for others.
- **Fairness**: Focus on equality or injustice.
- **Loyalty**: Rooted in patriotism or community solidarity.

- **Authority**: Respect for or criticism of authority figures.
- **Purity**: Relating to cultural or religious sanctity.

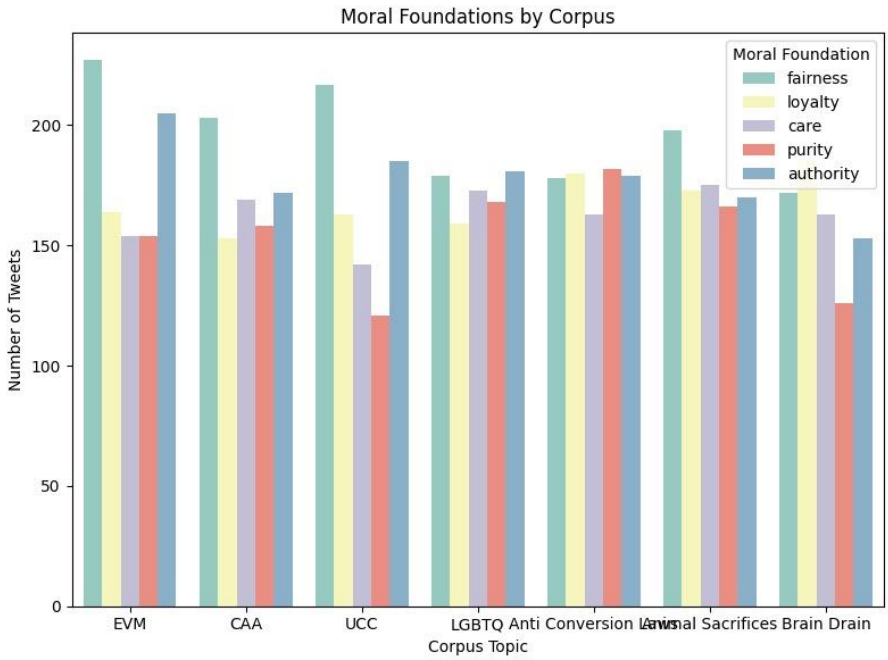

Fig 4.2.4. Moral Foundation Distribution by Corpus Topic

The graph illustrates the distribution of tweets with moral foundations across seven socio-political themes based on the Indian Twitter dataset. The study involves five moral foundations of Fairness, Loyalty, Care, Purity, and Authority based on the principles of Moral Foundations Theory.

Among analysis of different topics, Fairness is the most frequently used principle, which is most visibly seen in tweets on topics like Electronic Voting Machines (EVM) and the Uniform Civil Code (UCC). This indicates that the discussion in these topics is generally concerned with issues of justice, equity, and inclusive rule.

Tweets about LGBTQ rights and anti-conversion bills have a more balanced set of values for care, purity, and fairness.

On the other hand, the issue of animal sacrifices points towards a strong emphasis on the Purity dimension. The case of brain-drain, however, has a strong emphasis on the Authority and Fairness dimensions.

This moral foundations-based analysis provides valuable insights into value-based reasoning in online discourse. It points out that every socio-political problem in the Indian context evokes a distinct configuration of moral emotions, thereby underlining the capability of moral tagging frameworks to enhance our understanding and categorization of public opinion and online narratives.

5) Feature Extraction Using BERT:

- **Pre-trained Model Utilization:**
  A pre-trained **BERT base-uncased** model is employed to extract contextual embeddings for tweets. This transformer-based model captures deep semantic and syntactic relationships within the text.
- **Tokenization and Embedding Generation:**
  Each tweet is tokenized using BERT's tokenizer, which splits text into sub word units. The tokenized sequences are passed through the BERT model to generate dense, contextualized embeddings.
- **Feature Vector Representation:**
  The embedding corresponding to the [CLS] token is extracted to represent the overall meaning of the tweet. This [CLS] vector serves as the input feature to downstream classifiers for sentiment and moral foundation analysis.

6) Feature Extraction Using Bi-LSTM with TF-IDF:

- **TF-IDF Vectorization:**
  Textual data is vectorized using the **Term Frequency-Inverse Document Frequency (TF-IDF)** method, which converts tweets into sparse matrices based on the relative importance of each word in the corpus.
- **Sequence Modelling via Bi-LSTM:**
  The TF-IDF vectors are input to a **Bidirectional Long Short-Term Memory (Bi-LSTM)** network. This model captures sequential dependencies and bidirectional context within the tweet, allowing it to model the structure of text more effectively than traditional feed-forward networks.
- **Embedding Purpose:**
  The Bi-LSTM processes the TF-IDF inputs and produces high-level sequence representations, which are then used for classification in the subsequent fully connected layers.

7) Feature Extraction Using Bi-directional GRU with GloVe Embeddings:

- **Pre-trained Embedding Integration:**
  Pre-trained GloVe embeddings (e.g., 100-dimensional vectors trained on Twitter data) are used to initialize the embedding layer.

These embeddings capture semantic similarity and word-level context from large-scale corpora.

- **Tokenization and Embedding Lookup:**
  Each tweet is tokenized into individual words. These tokens are mapped to their corresponding GloVe vectors, forming a sequence of dense, fixed-size word embeddings representing the tweet.
- **Sequential Context Encoding:**
  The embedded word sequences are passed through a Bi-directional GRU network, which processes information in both forward and backward directions. This allows the model to capture contextual dependencies and semantic flow in informal and noisy Twitter text.
- **Feature Vector Representation:**
  The final hidden states from both GRU directions are concatenated to form a comprehensive representation of the tweet. This output vector serves as the feature input to downstream classifiers for sentiment and moral foundation analysis.

8) Feature Extraction Using GRU with TF-IDF Embeddings:

- **TF-IDF Vectorization of Text:**
  Each tweet is preprocessed (tokenized, lowercased, stopwords removed) and transformed into a sparse high-dimensional vector using the Term Frequency-Inverse Document Frequency (TF-IDF) method. This captures the importance of each word relative to the corpus, emphasizing rare but informative terms.
- **Fixed-Length Input Representation:**
  The TF-IDF vectors, though sparse, are converted into fixed-length input arrays. These serve as direct representations of word importance without relying on pre-trained semantic embeddings.
- **Dimensional Reduction (Optional):**
  To reduce computational complexity and mitigate sparsity, dimensionality reduction techniques such as PCA or Truncated SVD may be applied, producing denser, lower-dimensional input suitable for neural network processing.
- **Sequential Encoding via GRU:**
  The reduced feature vector (reshaped as a pseudo-sequence if required) is fed into a unidirectional GRU network. The GRU processes the sequence to identify abstract temporal patterns or feature transitions across weighted term indices.
- **Latent Feature Representation:**
  The final hidden state of the GRU serves as a compact, high-level representation of the tweet. This vector is then passed to the

classification layers for predicting sentiment polarity and moral foundation categories.

9) Model Evaluation:

- **Evaluation Metrics:**
  Model performance is assessed using standard classification metrics:
  - **Accuracy:** Overall correctness across all classes.
  - **Precision, Recall, F1-Score:** Computed for each class to evaluate performance on imbalanced data.
- **Validation Strategy:**
  K-fold cross-validation is used (with k=5) to ensure robustness, reduce variance in results, and avoid overfitting on the dataset.
- **Generalization Check:**
  Performance is also evaluated on a held-out test set to confirm the model's ability to generalize to unseen data.

10)Extensions and Improvements:

- **Hyperparameter Optimization:**
  Future work can integrate GridSearchCV or RandomizedSearchCV to tune model hyperparameters (e.g., learning rate, dropout rate, hidden dimensions) for BERT, Bi-GRU and Bi-LSTM pipelines.

11)Expected Outcomes:

- The Bi-LSTM model integrated with TF-IDF embeddings is expected to outperform Bi-GRU, particularly in moral foundation classification, owing to its enhanced capability in managing longer contextual information and more intricate sentence dependencies.
- Overall, this comparative analysis will emphasize the trade-offs between accuracy, computational demands, and model interpretability. While BERT is anticipated to excel in performance, Bi-LSTM will offer a balanced trade-off, and Bi-GRU will highlight the benefits of model efficiency in constrained environments. The TF-IDF model will illustrate the limitations of traditional approaches, further reinforcing the importance of contextual embeddings in modern NLP tasks.

4.3. Results and Analysis:

Table 4.1 Results of Sentiment Analysis

S.no	Encoding Type	Classifier/Model	Accuracy	Precision	Recall	F1-Score
1.	BERT	BERT	99.19	99.19	99.19	99.18
2.	TF-IDF	LSTM	78.24	78.36	78.24	78.25
3.	GloVe	Bi-directional GRU	72.93	74.33	72.93	72.91
4.	TF-IDF	GRU	70.01	70.79	70.01	70.02
5.	GloVe	LSTM	32.61	10.64	32.61	16.04

- These are the average values generated after 5 folds of running the respective models.
- These results indicate for sentiment analysis of the tweet data which determine the 3 different stances i.e. favor, against, neutral of the given topic.

Table 4.2 Results of Moral-Foundation Analysis

S.no	Encoding Type	Classifier/Model	Accuracy	Precision	Recall	F1-Score
1.	BERT	BERT	91.73	91.82	91.73	91.75
2.	TF-IDF	LSTM	62.44	62.75	62.44	62.50
3.	GloVe	Bi-directional GRU	51.67	53,10	51.67	51.42
4.	TF-IDF	GRU	45.73	47.33	45.73	45.16
5.	GloVe	LSTM	28.34	10.23	28.34	11.56

- These are the average values generated after 7 folds of running the respective model
- These results indicate the moral-foundation analysis of the tweet data which determine 5 different moral foundations i.e. care, fairness, loyalty, authority, purity.

4.3.1. Result Description for Sentiment Analysis:

```
Misclassified Tweets:

Final Results Across 5 Folds:
Avg Loss: 0.0336
Avg Accuracy: 0.9919
Avg Precision: 0.9919
Avg Recall: 0.9919
Avg F1-Score: 0.9918
```

Fig 4.3.1 Results for Sentiment Analysis using BERT Model

- The BERT-based sentiment analysis model achieved an average accuracy, precision, and recall of **99.19%** across 5-fold cross-validation. The low average loss of 0.0336 and a high F1-score of **99.18%** indicate that the model is highly effective and reliable for sentiment classification tasks in this context.

```
Final Results Across 5 Folds:
Avg Accuracy: 0.7824
Avg Precision: 0.7836
Avg Recall: 0.7824
Avg F1-Score: 0.7825
```

Fig 4.3.2 Results for Sentiment Analysis using Bi-LSTM Model

- The Bi-LSTM model achieved moderate performance on Indian based Twitter Dataset sentiment analysis, with an average accuracy of **78.24%** and an F1-score of **78.25%** across 5-fold cross-validation. While effective, its performance is lower compared to more advanced models like BERT.

```
Final Results Across 7 Folds:
Avg Accuracy: 0.7293
Avg Precision: 0.7433
Avg Recall: 0.7293
Avg F1-Score: 0.7291
```

Fig 4.3.3 Results for Sentiment Analysis using Bi-GRU + GloVe Model

- The Bi-GRU model with GloVe embeddings achieved moderate performance on sentiment analysis using the Indian Twitter dataset, with an average accuracy of 72.93% and an F1-score of 72.91% across 7-fold cross-validation. While the model demonstrated a solid balance between precision (74.33%) and recall (72.93%), its performance was slightly lower than that of the Bi-LSTM model and significantly below transformer-based approaches like BERT.

```
Final Results Across 7 Folds:
Avg Accuracy: 0.7001
Avg Precision: 0.7079
Avg Recall: 0.7001
Avg F1-Score: 0.7002
```

Fig 4.3.4 Results for Sentiment Analysis using Bi-GRU + TF-IDF Model

- The Bi-GRU model combined with TF-IDF embeddings for sentiment analysis demonstrated a consistent and moderately strong performance across 7-fold cross-validation. It achieved an average accuracy of 70.01%, with a precision of 70.79%, recall of 70.01%, and an F1-score of 70.02%. These metrics indicate that the model was able to maintain a balanced trade-off between correctly identifying relevant sentiments and minimizing false predictions

```
Final Results Across 7 Folds:
Avg Loss: 0.3752
Avg Accuracy: 0.9173
Avg Precision: 0.9182
Avg Recall: 0.9173
Avg F1-Score: 0.9175
```

Fig 4.3.5 Results for Moral-Foundation Analysis using BERT Model

- The BERT-based model achieved strong performance in classifying tweets according to the five moral foundations—Care, Fairness, Authority, Loyalty, and Purity—within the Indian Twitter context. With an average accuracy of **91.73%** and F1-score of **91.75%** across 7 folds, the model demonstrates high reliability and consistency. Additionally, some misclassified tweets are listed, reflecting challenges in nuanced or context-sensitive moral framing, such as gender identity or religious rituals. Despite this, the overall metrics indicate that the BERT model is highly effective for automated moral foundation detection in the Indian social media context.

```
Final Results Across 7 Folds:
Avg Accuracy: 0.6244
Avg Precision: 0.6275
Avg Recall: 0.6244
Avg F1-Score: 0.6250
```

Fig 4.3.6 Results for Moral-Foundation Analysis using Bi-LSTM Model

- The Bi-LSTM model was applied to classify Indian Twitter data into five moral foundations—Care, Fairness, Authority, Loyalty, and Purity. Evaluated over 7-fold cross-validation, the model achieved:
- Average Accuracy: 62.44%
- Average Precision: 62.75%
- Average Recall: 62.44%
- Average F1-Score: 62.50%

These results indicate moderate performance, with the model showing a relatively balanced precision and recall. While the Bi-LSTM captures general patterns in the moral foundation categories, it performs notably lower than the BERT-based approach, suggesting limitations in capturing the nuanced linguistic and contextual features present in moral discourse on Indian Twitter.

```
Final Results Across 7 Folds:
Avg Accuracy: 0.5167
Avg Precision: 0.5310
Avg Recall: 0.5167
Avg F1-Score: 0.5142
```

Fig 4.3.7 Results for Moral-Foundation Analysis using GRU Model

- The figure demonstrates the average performance metrics over 7-fold cross-validation with a GRU model to classify tweets based on moral foundations (e.g., care, fairness, loyalty, authority, purity). The model was evaluated on the ability to detect moral framing in text.

Performance Indicators:

- Mean Accuracy: 0.5167%
- Mean Precision: 0.5310%
- Mean Recall: 0.5167%
- Mean F1-Score: 0.5142%

These scores reflect moderate classification ability. The fairly low measures imply that although the model does pick up on some moral sentiment, it will have difficulty with overlap among moral categories

```
Final Results Across 7 Folds:
Avg Accuracy: 0.4573
Avg Precision: 0.4733
Avg Recall: 0.4573
Avg F1-Score: 0.4516
```

Fig 4.3.8 Results for Moral-Foundation Analysis using GRU + TF-IDF Model

- The Bi-GRU model with TF-IDF embeddings, when applied to moral foundation analysis, showed relatively lower performance across 7-fold cross-validation. It achieved an average accuracy of 45.73%, with precision at 47.33%, recall at 45.73%, and an F1-score of 45.16%. These results indicate that while the model captured some relevant patterns, it struggled to effectively classify moral foundations, likely due to the limited contextual representation of TF-IDF and the simpler architecture of GRU.

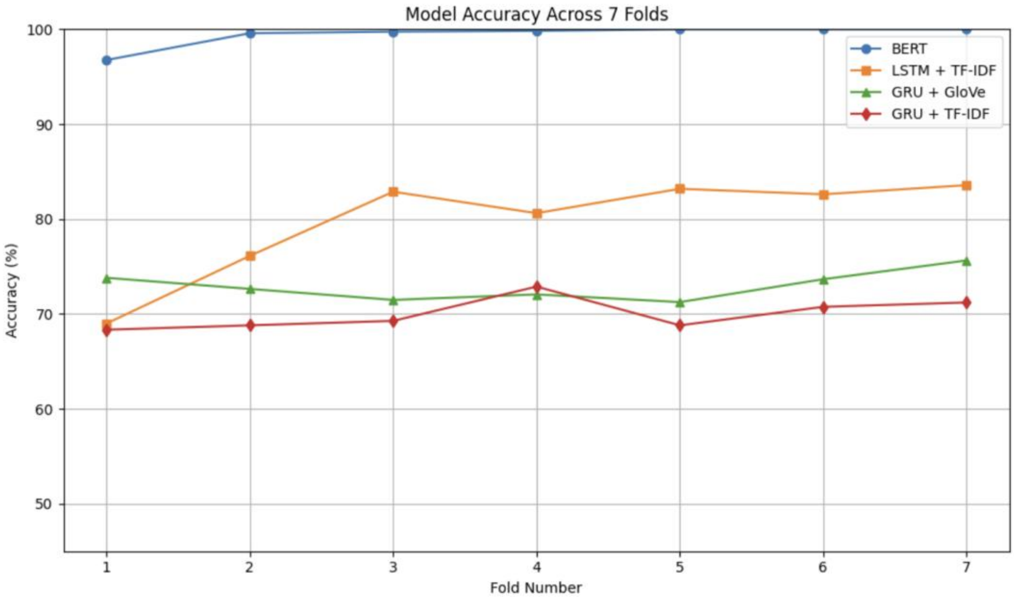

Fig 4.3.9 Model Accuracy Across 7 Folds for Sentiment Analysis

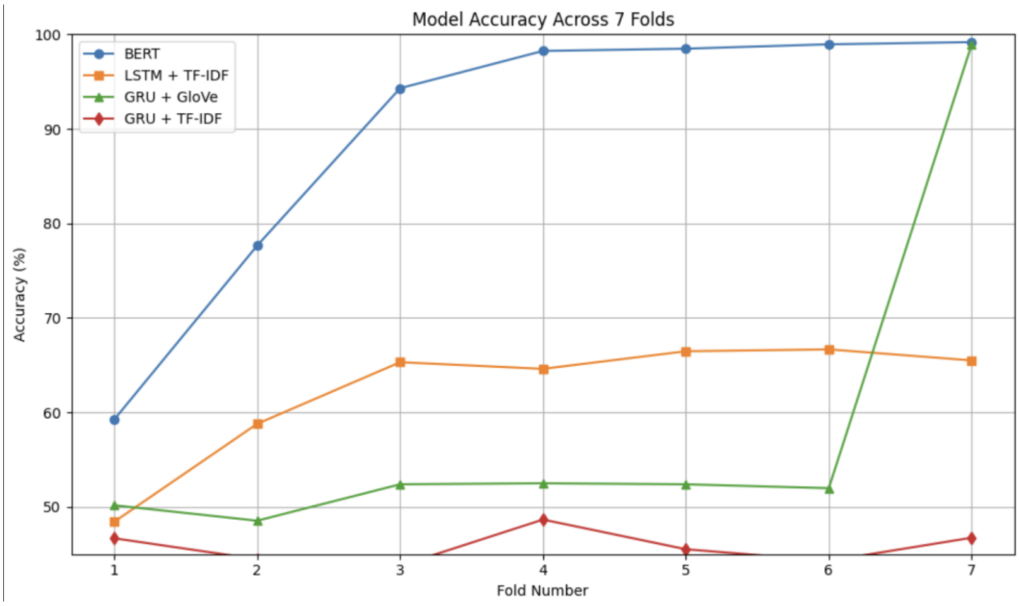

Fig 4.4 Model Accuracy Across 7 Folds for Moral Foundations

5. CONCLUSIONS AND FUTURE SCOPE

Conclusion:

Concluding our project, the morality and stance classification framework developed marks a significant step forward in the automated understanding of user sentiment and moral positioning within Twitter discourse—particularly within the Indian sociopolitical context.

While the majority of recent sentiment and morality classification models have been trained and tested on global datasets, including Twitter data, there exists a noticeable gap in research focusing on Indian-specific sociopolitical content. This work addresses that gap by constructing and analyzing a novel dataset of 6,011 tweets related to issues such as the Citizenship Amendment Act (CAA), Uniform Civil Code (UCC), LGBTQ+ rights, etc., thereby offering a more context-aware, culture-sensitive benchmark for sentiment and morality classification.

Our experimental evaluation shows that BERT significantly outperformed the other models with accuracy scores of 99.19% for sentiment and 91.73% for morality, owing to its deep contextual embeddings. Bi-LSTM models showed moderate performance, achieving 78.24% for sentiment and 62.44% for morality. Although Bi-GRU models recorded slightly lower accuracies (71.93% and 52.67% respectively), they proved to be lightweight and efficient, making them suitable for scenarios with limited computational resources, whereas GRU model with TF-IDF encodings have about 70.01% and 45.73% respectively. Finally, the implantation of LSTM with GloVe embeddings proves with the least readings of all the working models with accuracy of 32.61% and 28.34% respectively.

Overall, this project highlights the effectiveness of transformer-based models while acknowledging the practical utility of recurrent neural networks in understanding diverse, culturally rich, and low-resource settings like India. It sets a foundation for real-time social media analysis tools capable of detecting public sentiment and moral perspectives with high precision.

Future-Scope:

In the next phase of development, the framework can benefit significantly from expanding the dataset—both in size and diversity. Including a broader range of sociopolitical topics, incorporating multilingual data (such as Hindi, Tamil, Bengali, etc.), and addressing code-mixed language (commonly used in Indian tweets) will enhance the model’s generalizability and cultural relevance. While

BERT remains the most accurate, lighter models like Bi-GRU and Bi-LSTM can be further improved through the use of attention mechanisms, hybrid architectures, and transfer learning to balance performance with computational efficiency. The inclusion of platform-specific features such as hashtags, retweets, mentions, emojis, and even user metadata (like follower count or verified status) can enrich the input representation, allowing the models to make more nuanced predictions.

Moreover, building real-time processing capabilities is a critical goal for scaling this framework to handle large-scale live Twitter data. Real-time sentiment and morality classification would be invaluable for use cases such as social media monitoring, content moderation, and early detection of toxic trends or misinformation. An open-source API could be developed to allow integration with external applications—enabling developers, researchers, and policymakers to leverage the model for public opinion tracking, trend forecasting, or digital governance tools. Furthermore, adapting the framework to work across multiple platforms (e.g., Facebook, Instagram, YouTube comments) and regional languages will make it more inclusive and versatile. Ultimately, this system has the potential to contribute meaningfully to ethical AI development, digital civility, and data-driven decision-making in public discourse analysis.

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APPENDIX

Open Access:

https://github.com/nlvmadhav/moralFoundations.git

Code:

- **Moral-foundation analysis**

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import random

import numpy as np

from collections import Counter

import re

import torch.nn as nn

from sklearn.preprocessing import MultiLabelBinarizer

from transformers import pipeline, BertTokenizer, BertForSequenceClassification,
Trainer, TrainingArguments,BertPreTrainedModel, BertConfig,BertModel,
AutoTokenizer,Trainer, TrainingArguments, AdamW, get_scheduler

import torch

from datasets import Dataset

from sklearn.metrics import accuracy_score, f1_score,precision_score,
recall_score,confusion_matrix,ConfusionMatrixDisplay

from sklearn.model_selection import train_test_split

print(torch.cuda.is_available())

data = pd.read_csv(r'C:\Users\anura\OneDrive\Desktop\proj\updated_data.csv',
encoding= "ISO-8859-1")

data.head()

class_counts = data.iloc[:, 0].value_counts()


print("Number of rows for each class:")
```

```
print(class_counts)
total_rows = data.shape[0]
print("\nTotal number of rows:", total_rows)
plt.figure(figsize=(8, 6))
sns.countplot(x='Corpus', data=data)
plt.title('Distribution of Classes', fontsize=16)
plt.xlabel('Corpus', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.tight_layout()
plt.show()
def clean_tweet(tweet):
    tweet = re.sub(r'@[\w_]+', '', tweet)
    tweet = re.sub(r'#\w+', '', tweet)
    tweet = re.sub(r'http\S+|www\S+', '', tweet)
    tweet = re.sub(r'[^A-Za-z0-9\s]+', '', tweet)
    tweet = tweet.strip().lower()
    return tweet
data['Tweet Text'] = data['Tweet Text'].apply(clean_tweet)
data.head()


moral_foundations = ["Care", "Fairness", "Loyalty", "Authority", "Liberty"]
#data seg
zero_samples = data.sample(frac=0.50, random_state=55)
# Select threshold for final multi-label classification
final_threshold = 0.40
all_predicted_labels = []
d = dict()
```

```
scores = []

for text in zero_samples["Tweet Text"]:

    result = classifier(text, moral_foundations)

    print(result)

    scores.append(result['scores'])

    # Append selected labels

    selected_labels = [label for label, score in zip(result["labels"], result["scores"]) if
score >= final_threshold]

    d[text] = tuple(selected_labels)

    all_predicted_labels.extend(selected_labels)


# Count label occurrences

label_counts = Counter(all_predicted_labels)

print(scores)

len(all_predicted_labels)

len(zero_samples["Tweet Text"])

d = {k: v for k, v in d.items() if len(d[k]) != 0}

print(d)

len(d)
```

Loading Zero Shot data

- Zero shot classification is done using chatgpt llm

- Classified about 50% of the total tweets

```
data1 = pd.read_csv("Few_shot.csv")

data1.head()

data1 = data[data["Foundation"].notna()]

from collections import Counter


d = dict()  # Dictionary to store tweet text and selected labels
```

```
all_predicted_labels = []  # List to collect all predicted labels

# Loop through each Tweet Text
for text in data1["Tweet Text"]:
    # Assuming you have a classifier that returns a list of selected labels for each text
    # result = classifier(text, moral_foundations)  # Uncomment if you have a classifier

    # Simulating selected labels (replace this with actual logic)
    selected_labels = data1.loc[data1["Tweet Text"] == text, "Foundation"].tolist()

    # Store in dictionary
    d[text] = tuple(selected_labels)
    # Collect all labels for counting
    all_predicted_labels.extend(selected_labels)

# Count occurrences of each label
label_counts = Counter(all_predicted_labels)

# Display results
print(d)  # Dictionary with text → labels
print(label_counts)  # Count of each foundation label
# from transformers import pipeline

# # Load the zero-shot classification pipeline
# classifier = pipeline("zero-shot-classification", model="facebook/bart-large-mnli")

# text = "Helping the poor is a moral duty."
```

```
# candidate_labels = ["Care", "Fairness", "Loyalty", "Authority", "Sanctity",
"Liberty"]

# # Perform zero-shot classification

# result = classifier(text, candidate_labels)

# print(result)

## Few Shot learing

###     -Creating a data Farme of Zero shot labels and feeding them to compute few
shot prediction

for item in d:

    d[item] = d[item][0].split(', ')

len(d)

#data frame creation

few_shot_data = pd.DataFrame({

    "text": d.keys(),

    "labels":d.values()

})

few_shot_data.head()

train_dataset, test_dataset = train_test_split(

    few_shot_data, test_size=0.2, random_state=42, shuffle=True

)

#creating encodings

mlb = MultiLabelBinarizer()

train_label = mlb.fit_transform(train_dataset['labels'])

test_label = mlb.fit_transform(test_dataset['labels'])

print(train_label)

moral_foundations = list(mlb.classes_)

print(moral_foundations)

#tokenizing the text

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
```

```
train_inputs = tokenizer(list(train_dataset["text"]), padding = True, truncation = True,
return_tensors = "pt")

test_inputs = tokenizer(list(test_dataset["text"]), padding = True, truncation = True,
return_tensors = "pt")

### Creating a data pipeline

class BertForMultiLabelClassification(BertPreTrainedModel):

  def __init__(self, config):

    super().__init__(config)

    self.bert = BertModel(config)

    for param in self.bert.parameters():

      param.requires_grad = False

    for param in self.bert.encoder.layer[-7:].parameters():  # Unfreeze last 2 layers

      param.requires_grad = True

    self.dropout = nn.Dropout(0.2)  # Prevent overfitting

    # self.fc1 = nn.Linear(config.hidden_size, 512)

    # Fully connected classifier

    self.classifier = nn.Linear(config.hidden_size, config.num_labels)  # Multi-label
output

  def forward(self, input_ids, attention_mask=None, token_type_ids=None,
labels=None):

    outputs = self.bert(input_ids, attention_mask=attention_mask,
token_type_ids=token_type_ids, )

    pooled_output = outputs.pooler_output  # [CLS] token representation

    pooled_output = self.dropout(pooled_output)

    # pooled_output = self.fc1(pooled_output)

    logits = self.classifier(pooled_output)  # Output logits


    loss = None

    if labels is not None:
```

```
            loss_fct = nn.CrossEntropyLoss(label_smoothing=0.1)  # Multi-label
classification loss

            loss = loss_fct(logits, labels)  # Ensure labels are float for BCE

        return {"loss": loss, "logits": logits}

num_labels = len(moral_foundations)  # Number of labels

config = BertConfig.from_pretrained("bert-base-uncased", num_labels=num_labels)

model = BertForMultiLabelClassification.from_pretrained("bert-base-uncased",
config=config)

model.to('cuda')

#labels to torch tensors

train_labels = torch.tensor(train_label, dtype = torch.float)

test_labels = torch.tensor(test_label, dtype = torch.float)

training_args = TrainingArguments(

    output_dir="./results",

    eval_strategy="epoch",

    save_strategy="epoch",

    save_total_limit=2,

    load_best_model_at_end=True,

    metric_for_best_model="f1",

    greater_is_better=True,

    per_device_train_batch_size=4,  # Increased batch size

    per_device_eval_batch_size=4,

    num_train_epochs=5,  # Reduce to avoid overfitting

    warmup_steps=100,  # Stabilize training

    weight_decay=0.01,

    logging_dir="./logs",

    logging_steps=10,

    report_to="none",

    fp16=True,  # Use mixed precision
```

```
    gradient_accumulation_steps=4,  # Effective batch size of 32
    learning_rate=2e-5,  # More stable LR
    #lr_scheduler_type="cosine",  # Better decay
    seed=42
)
#dataset
class TweetData(torch.utils.data.Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        if len(labels.shape) > 1 and labels.shape[1] > 1:
            labels = labels.argmax(axis=1)
        self.labels = labels
#initializing the data set
train_dataset = TweetData(train_inputs, train_labels)
test_dataset = TweetData(test_inputs, test_labels)
def compute_metrics(eval_pred):
    logits, labels = eval_pred
        # Convert logits to predicted class indices
    predictions = logits.argmax(axis=1)
    # Convert one-hot encoded labels to class indices (if needed)
    if len(labels.shape) > 1 and labels.shape[1] > 1:
        labels = labels.argmax(axis=1)
    acc = accuracy_score(labels, predictions)
    f1 = f1_score(labels, predictions, average="weighted")
    precision = precision_score(labels, predictions, average="weighted")
    recall = recall_score(labels, predictions, average="weighted")

    return {
```

```
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }
optimizer = AdamW([
    {"params": model.bert.encoder.layer[-7:].parameters(), "lr": 1e-5},  # Fine-tune last 6 layers
    # {"params": model.fc1.parameters(), "lr": 2e-3},  # Higher LR for classifier
    {"params": model.classifier.parameters(), "lr": 1e-3}
], weight_decay=0.01)
num_training_steps = len(train_dataset) * training_args.num_train_epochs  # Total training steps
lr_scheduler = get_scheduler(
    name="linear",
    optimizer=optimizer,
    num_warmup_steps=0,
    num_training_steps=num_training_steps
)
#setting the trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics,
    optimizers = (optimizer,lr_scheduler)
)
trainer.train()
```

```
results = trainer.evaluate()
print("Evaluation results:", results)
# Move input tensors to GPU
test_inputs = {k: v.to('cuda') for k, v in test_inputs.items()}
# Store predictions
all_predictions = []
batch_size = 32
# Predict in batches
model.eval()  # Set model to evaluation mode
with torch.no_grad():  # Disable gradient calculation for inference
    for i in range(0, len(remain_tweets), batch_size):
        batch_inputs = {k: v[i:i + batch_size] for k, v in test_inputs.items()}
        batch_outputs = model(**batch_inputs)  # Get outputss
        batch_logits = batch_outputs["logits"]
        batch_probs = torch.sigmoid(batch_logits)  # Convert to probabilities
len(external_data)
external_data.head()
from collections import Counter

d1 = dict()  # Dictionary to store tweet text and selected labels
all_predicted_labels = []  # List to collect all predicted labels

# Loop through each Tweet Text
for text in external_data["Tweet Text"]:
    # Assuming you have a classifier that returns a list of selected labels for each text
    # result = classifier(text, moral_foundations)  # Uncomment if you have a classifier
    # Simulating selected labels (replace this with actual logic)
```

selected_labels = external_data.loc[external_data["Tweet Text"] == text, "Foundations"].tolist()

- **Sentiment Analysis**

Sentiment analysis for Twitter Text

- Primary step of the Morality based sentimental analysis. Text is tockenized using TF-IDF vectorization and BERT

Load dataset from CSV file

file_path = "/content/updated_data.csv" # Replace with the actual file path

df = pd.read_csv(file_path, encoding = "ISO-8859-1")

Display the first few rows of the dataset

print(df.head())

Find the number of rows for each unique value in the first column

class_counts = df.iloc[:, 0].value_counts()

Print the counts for each class

print("Number of rows for each class:")

print(class_counts)

Print the total number of rows

total_rows = df.shape[0]

print("\nTotal number of rows:", total_rows)

Preprocess the data

X = df['Tweet Text']

y = df['Stance']+1

Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

Bert Model

- The data set is limited(1213 Tweets).

- Model is designed with bert as encoder and fully connected layers as classifier

- Performing K(5 here) fold cross validation for analyzing the performance of the model on the dataset.

```
# Model Setup

class CustomDataset(Dataset):

    def __init__(self, texts, labels, tokenizer, max_len):

        self.texts = texts

        self.labels = labels

        self.tokenizer = tokenizer

        self.max_len = max_len


    def __len__(self):

        return len(self.texts)


    def __getitem__(self, item):

        text = str(self.texts[item])

        label = self.labels[item]

        encoding = self.tokenizer.encode_plus(

            text,

            add_special_tokens=True,

            max_length=self.max_len,

            padding='max_length',

            truncation=True,

            return_tensors='pt'

        )

        return {

            'input_ids': encoding['input_ids'].flatten(),

            'attention_mask': encoding['attention_mask'].flatten(),
```

```
            'label': torch.tensor(label, dtype=torch.long)
        }
class BertForSentimentAnalysis(BertPreTrainedModel):
    def __init__(self, config):
        super().__init__(config)
        self.bert = BertModel(config)
        self.dropout = nn.Dropout(0.3)  # Reduce dropout

        # self.fc1 = nn.Linear(config.hidden_size, 512)  # One small FC layer
        # self.relu = nn.ReLU()
        self.classifier = nn.Linear(config.hidden_size, 3)  # Output layer
    def forward(self, input_ids, attention_mask=None, token_type_ids=None,
labels=None):
        outputs = self.bert(input_ids, attention_mask=attention_mask,
token_type_ids=token_type_ids)
        pooled_output = outputs.pooler_output  # [CLS] token representation
        x = self.dropout(pooled_output)
        # x = self.fc1(x)
        # x = self.relu(x)
        logits = self.classifier(x)  # Final predictions
        loss = None
        if labels is not None:
            loss_fct = nn.CrossEntropyLoss()
            loss = loss_fct(logits, labels)
        return {"loss": loss, "logits": logits}
def k_fold_cross_validation(model, dataset, texts, k=5, batch_size=8, epochs=3,
lr=2e-5):
    kf = KFold(n_splits=k, shuffle=True, random_state=42)
```

```
    losses, accuracies, precisions, recalls, f1_scores = [], [], [], [], []

    for fold, (train_idx, val_idx) in enumerate(kf.split(dataset)):
        print(f"\n� Fold {fold+1}/{k}")

        # Create data samplers
        train_sampler = SubsetRandomSampler(train_idx)
        val_sampler = SubsetRandomSampler(val_idx)

        train_loader = DataLoader(dataset, batch_size=batch_size,
sampler=train_sampler)
        val_loader = DataLoader(dataset, batch_size=batch_size, sampler=val_sampler)

        trainer = Trainer(
            model=model,
            args=training_args,
            train_dataset=dataset,
            eval_dataset=dataset,  # Full dataset, but using samplers
        )
        # Train & Evaluate
        trainer.train()
        eval_result = trainer.evaluate()
        losses.append(eval_result["eval_loss"])

        # Store predictions & labels
        all_preds, all_labels, misclassified_samples = [], [], []

        model.eval()
```

```
        with torch.no_grad():
            for batch_idx, batch in enumerate(val_loader):
                input_ids = batch["input_ids"].to(device)
                attention_mask = batch["attention_mask"].to(device)
                labels = batch["label"].to(device)
                outputs = model(input_ids, attention_mask=attention_mask)
                logits = outputs["logits"]
                preds = torch.argmax(logits, dim=1)
                all_preds.extend(preds.cpu().numpy())
                all_labels.extend(labels.cpu().numpy())

                # Store misclassified samples
                for i, (pred, true) in enumerate(zip(preds, labels)):
                    if pred != true:
                        misclassified_samples.append((texts[val_idx[batch_idx * batch_size + i]], true.item(), pred.item()))


bert_tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
MAX_LEN = 128
bert_train_dataset = CustomDataset(X_train.tolist(), y_train.tolist(), bert_tokenizer, MAX_LEN)
bert_test_dataset = CustomDataset(X_test.tolist(), y_test.tolist(), bert_tokenizer, MAX_LEN)
bert_labels, bert_preds = [],[]
def evaluate_on_test(model, test_dataset, batch_size=8):
    test_loader = DataLoader(test_dataset, batch_size=batch_size)
    # Store predictions & true labels
    model.eval()  # Set model to evaluation mode
    with torch.no_grad():
```

```
        for batch in test_loader:
            input_ids = batch["input_ids"].to(device)
            attention_mask = batch["attention_mask"].to(device)
            labels = batch["label"].to(device)

            outputs = model(input_ids, attention_mask=attention_mask)
            logits = outputs["logits"]
            preds = torch.argmax(logits, dim=1)

            bert_preds.extend(preds.cpu().numpy())
            bert_labels.extend(labels.cpu().numpy())


config = BertConfig.from_pretrained("bert-base-uncased", num_labels=3)
model = BertForSentimentAnalysis.from_pretrained("bert-base-uncased",
config=config)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
text = df["Tweet Text"].tolist()
# Perform K-Fold Cross-Validation on training data
k_fold_cross_validation(model, bert_train_dataset, k=5, batch_size=8, epochs=3,
lr=2e-5, texts = text)
evaluate_on_test(model, bert_test_dataset, batch_size=8)
# Tfidf with lstm
- The data set is limited(1207 Tweets).
- Data encoded using TF-IDF vectorizer and fed to lstm for classification
- Performing K(5 here) fold cross validation for analyzing the model
class SentimentDataset(Dataset):
    def __init__(self, sequences, labels, vectorizer):
```

```
        self.sequences = vectorizer.transform(sequences).toarray()  # Transform text to numerical features
        self.labels = labels
    def __len__(self):
        return len(self.labels)
    def __getitem__(self, idx):
        return {
            'features': torch.tensor(self.sequences[idx], dtype=torch.float32),
            'label': torch.tensor(self.labels[idx], dtype=torch.long)
        }
class LSTMClassifier(nn.Module):
    def __init__(self, input_dim, hidden_dim, output_dim):
        super(LSTMClassifier, self).__init__()
        self.lstm = nn.LSTM(input_dim, hidden_dim, batch_first=True)
        self.hidden1 = nn.Linear(hidden_dim, 128)
        self.hidden2 = nn.Linear(128, 64)
        self.hidden3 = nn.Linear(64, 32)
        self.classifier = nn.Linear(32, output_dim)
        self.dropout = nn.Dropout(0.3)
    def forward(self, x):
       # Reshape input to (batch_size, sequence_length=1, input_size)
       x = x.unsqueeze(1)
       _, (hidden, _) = self.lstm(x)
       output = self.dropout(hidden[-1])
       output = self.hidden1(output)
       output = self.dropout(output)
       output = self.hidden2(output)
       output = self.dropout(output)
```

```
        output = self.hidden3(output)
        output = self.dropout(output)
        return self.classifier(output)
def k_fold_cross_validation(model, dataset, texts, labels, k=5, batch_size=8,
epochs=5, lr=2e-5):
    kf = KFold(n_splits=k, shuffle=True, random_state=42)
    class_weights = compute_class_weight(class_weight='balanced',
classes=np.unique(labels), y=labels)
    class_weights = torch.tensor(class_weights, dtype=torch.float32).to(device)
    criterion = nn.CrossEntropyLoss(weight=class_weights)
    optimizer = optim.AdamW(model.parameters(), lr=lr)
    scheduler = optim.lr_scheduler.StepLR(optimizer, step_size=2, gamma=0.5)
    accuracies, precisions, recalls, f1_scores = [], [], [], []
        train_sampler = SubsetRandomSampler(train_idx)
        val_sampler = SubsetRandomSampler(val_idx)

        train_loader = DataLoader(dataset, batch_size=batch_size,
sampler=train_sampler)
        val_loader = DataLoader(dataset, batch_size=batch_size, sampler=val_sampler)

        for epoch in range(epochs):
            model.train()
            for batch in train_loader:
                optimizer.zero_grad()
                features = batch['features'].to(device)
                labels = batch['label'].to(device)
                outputs = model(features)
                loss = criterion(outputs, labels)
                loss.backward()
```

```
            optimizer.step()
        scheduler.step()
    model.eval()
    all_preds, all_labels = [], []
    with torch.no_grad():
        for batch in val_loader:
            features = batch['features'].to(device)
            outputs = model(features)
            preds = torch.argmax(outputs, dim=1)
            all_preds.extend(preds.cpu().numpy())
            all_labels.extend(batch['label'].cpu().numpy())
    accuracy = accuracy_score(all_labels, all_preds)
    precision, recall, f1, _ = precision_recall_fscore_support(all_labels, all_preds,
average="weighted", zero_division=0)
vectorizer = TfidfVectorizer(max_features=512)
features = vectorizer.fit_transform(X).toarray()
vocab = vectorizer.get_feature_names_out()
input_dim = features.shape[1]
hidden_dim = 256
output_dim = 3
print(input_dim)
dataset = SentimentDataset(X, y, vectorizer)
model = LSTMClassifier(input_dim, hidden_dim, output_dim)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
k_fold_cross_validation(model, dataset, X, y)
```

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Morality and Sentiment Analysis of Indian Twitter Discourse Using Deep Learning

Anurag Sai Y[1], Ashish Sadhu[1], Madhav NLV[1], Dharma Reddy R[1]

Abstract

The understanding of ethical viewpoints and sentiments in online conversation is a complex task, especially in culturally diverse and dynamic environments such as India. Although the majority of the existing models have been benchmarked on global or Western-centric(5)(10) data, the most significant gap remains in assessing the performance of models on Indian Twitter data. This research aims to bridge this gap by introducing a tailored dataset of 6,011 tweets centered on modern sociopolitical topics in India—thus providing a more culturally nuanced and realistic testing ground. We compare the performance of Bi-LSTM models with TF-IDF embeddings and the transformer-based BERT-base-uncased model for sentiment and moral foundation classification tasks. We also compare the performance of more compact recurrent models, namely Bi-GRU with GloVe embeddings and Bi-GRU with TF-IDF embeddings. Applying the k-fold cross-validation method, BERT shows excellent performance—achieving 99.19% in sentiment analysis and 91.73% in moral foundation classification—while Bi-LSTM achieves 78.24% and 62.44%, respectively. The Bi-GRU architecture, although compact and efficient, achieved moderate performance with scores of 71.93% in sentiment and 52.67% in moral classification, followed closely by the Bi-GRU + TF-IDF model, which achieved 70.01% and 45.73%, respectively.These results underscore the significance of contextually-aware embeddings in detecting subtle public sentiment and moral reasoning in a culturally particularized environment. This paper emphasizes the significance of localizing AI models to local contexts(India in this case) and provides valuable insights for the analysis of political behavior, digital governance, and the development of culturally-sensitive ethical AI systems.

Link: `https://github.com/nlvmadhav/moralFoundationsAnalysis.git`

Keywords

Sentiment analysis, Moral Foundation Theory(MFT), Indian Twitter data, BERT, Bi-LSTM, Deep learning, TF-IDF

Introduction

With over 400 million active users, Twitter has become a pivotal arena for public discourse and sociopolitical debate (9). Short, informal posts—often laced with cultural references and emotional cues—offer a unique window into collective opinion. Traditional sentiment analysis classifies text as positive, negative, or neutral, but fails to capture the deeper moral dimensions that guide human judgments (2). Moral Foundations Theory (MFT) (4) provides a psychological framework comprising five foundations: Care, Fairness, Loyalty, Authority, and Purity. Prior studies have applied MFT to Western social media datasets, yet the Indian sociocultural landscape, characterized by multilingualism, religious diversity, and political complexity, remains underexplored. This study aims to fill this gap by constructing an Indian-specific Twitter corpus and evaluating deep learning approaches for both sentiment and moral foundation classification.

Related Work

Early work in sentiment analysis laid the groundwork for opinion mining (7). Moral classification emerged through lexicon-based methods (12) and, more recently, transformer-based models. However, these efforts have been primarily Western-centric. In the Indian context, sentiment analysis has focused on regional languages (3), while moral reasoning remains largely uncharted. Our contribution extends these frameworks to analyze both sentiment and morality on Indian Twitter, providing a culturally grounded study.

Dataset Construction and Annotation

Dataset Construction

Currently, the majority of moral and sentiment analysis studies using Twitter data have been global in scope, with little attention devoted to explorations in the Indian socio-political context. To mitigate this and counter cultural specificity, it was necessary to construct a special data set representative of the new Indian moral discourse. In this paper, a data set of 6,011 tweets was constructed using the Twitter API, gathering information over a period of three months from

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January to March 2024. Tweets were pre-filtered using a handpicked set of hashtags—`#CAA`,`#UCC`, `#LGBTQ`,`#EVM`, `#AnimalSacrifice`, `#AntiConversion`, and `#BrainDrain` selected for high applicability to new Indian sociopolitical discourses.

The choice of these topics was based on two factors: first, to mirror the current controversies in the context of civil rights, religious freedoms, and reform policies; and second, to have a mix of ideological viewpoints, ranging from progressive to conservative thinking, especially in the context of modern India. They are topics like electoral integrity (`#EVM`), individual rights (`#LGBTQ`), and policy-based topics (`#BrainDrain`). By including both morally controversial and neutral topics, the dataset tries to capture a large range of opinions and ideological views.

Dataset Statistics

Table 1. Topic-wise Tweet Distribution

Topic	#Tweets	% of Corpus
CAA	855	14.2
UCC	828	13.8
LGBTQ	860	14.3
EVM	904	15.0
Animal Sacrifice	882	14.7
Anti-Conversion	882	14.7
Brain Drain	800	13.3

In Table 1, we present the distribution of tweets across various topics in the dataset, reflecting key social and political issues prevalent in India. Each topic represents significant areas of discourse that have shaped public opinion and national debates in the country. Below is a generalized description of each topic within the Indian context:

- **CAA (Citizenship Amendment Act)**: This topic, consisting of 855 tweets (14.2% of the dataset), addresses the Citizenship Amendment Act (CAA), a controversial piece of legislation passed by the Indian government. The Act provides a pathway to citizenship for refugees from neighboring countries, but its provisions have sparked significant debate regarding its implications for national identity, secularism, and minority rights. The topic captures the public discourse on the law's potential impact on the social fabric and its role in national politics, particularly during protests and public demonstrations.
- **UCC (Uniform Civil Code)**: Representing 13.8% of the dataset with 828 tweets, the Uniform Civil Code (UCC) is a proposed set of laws that seeks to replace personal laws governing marriage, inheritance, and adoption with a single unified code for all citizens. The topic is central to discussions about equality and social justice, focusing on the challenges of ensuring equal rights across diverse communities while respecting cultural and personal practices. The discourse around UCC reflects broader concerns about unity, gender equality, and the role of law in regulating personal matters.
- **LGBTQ**: With 860 tweets (14.3%), the LGBTQ topic captures the ongoing discussions around the rights of sexual minorities in India. The LGBTQ+ community has faced significant legal and social challenges, although recent years have seen important legal strides, including the decriminalization of same-sex relationships. Despite these advancements, the community continues to face social stigma, discrimination, and a lack of comprehensive legal protection. This topic is central to conversations about equality, social inclusion, and the recognition of diverse identities in Indian society.
- **EVM (Electronic Voting Machines)**: This topic, comprising 904 tweets (15.0%), focuses on the use of Electronic Voting Machines (EVMs) in India's electoral process. EVMs have been at the center of numerous debates and controversies, with various political groups questioning their reliability and security. Allegations of manipulation or malfunction in elections have fueled concerns about the integrity of India's democratic process. The topic reflects public concerns about transparency, electoral fairness, and the trustworthiness of technology used in national elections.
- **Animal Sacrifice**: The topic of animal sacrifice, with 882 tweets (14.7%), reflects ongoing discussions around the practice of sacrificing animals in certain traditional rituals. The practice has sparked debates about the ethical treatment of animals and the role of state intervention in regulating such activities. This issue touches on the balance between cultural practices and the protection of animal rights, as well as the broader discourse on the role of government in shaping ethical norms.
- **Anti-Conversion**: Also consisting of 882 tweets (14.7%), the Anti-Conversion topic deals with laws aimed at preventing fraudulent or coerced religious conversions. Several states in India have passed anti-conversion laws that restrict or regulate conversion activities, particularly those involving vulnerable individuals. The discourse around anti-conversion laws often centers on issues of freedom, autonomy, and the protection of individuals from exploitation, reflecting concerns about the impact of such laws on personal liberties and societal cohesion.
- **Brain Drain**: The Brain Drain topic, with 800 tweets (13.3%), captures discussions about the emigration of highly skilled professionals from India in search of better career opportunities, research facilities, and living conditions abroad. This phenomenon, commonly referred to as "brain drain," raises concerns about the loss of intellectual capital and the challenges faced by India in retaining its talent pool. The topic is connected to issues such as job opportunities, quality of education, and political stability, all of which influence the decision to migrate for career advancement.

Annotation Process

Each tweet in the dataset was annotated with two types of labels: sentiment and moral foundation. These dual annotations enable a comprehensive analysis of both the stance and the ethical framing of sociopolitical discourse on Indian Twitter.

Sentiment Labels

Tweets were classified into one of three sentiment categories: **favor**, **against**, or **neutral**. These categories reflect the author's attitude toward the sociopolitical issue referenced in the tweet. This tri-level sentiment scheme was designed to capture stance rather than emotional sentiment alone, offering a richer understanding of public alignment in polarized discussions. Sentiment assignment relied on the overall tone, linguistic cues, and contextual indicators found in the text. Examples of classification include:

- **Favor:** Tweets expressing support, agreement, or positive alignment with the topic.
- **Against:** Tweets showing opposition, disagreement, or negative sentiment.
- **Neutral:** Informational or observational tweets lacking explicit evaluative language.

To support effective model training, the dataset was curated to maintain as much balance across sentiment labels as possible. This sentiment framework serves as a complementary dimension to the moral foundation labels, allowing for a more layered understanding of digital opinion expression.

Moral Foundations Labels

Moral categorization follows the five-factor framework of Moral Foundations Theory (2), encompassing the following domains:

- **Care:** Protection and well-being of others.
- **Fairness:** Justice, equality, and reciprocity.
- **Loyalty:** Commitment to group, community, or nation.
- **Authority:** Respect for leadership, tradition, and hierarchy.
- **Purity:** Concerns about moral or physical cleanliness.

Each tweet was assigned a single dominant moral foundation label, determined by its primary ethical theme. Unlike some models that account for both virtues and vices, this study adopts a simplified scheme, assigning the most salient moral concern expressed in the tweet.

Figure 1 presents the distribution of stances across various socio-political topics within the corpus. Each topic comprises tweets categorized into one of three stance classes: **-1 (Opposing)**, **0 (Neutral)**, and **1 (Supporting)**. This visualization provides insight into how public sentiment varies by issue in Indian discourse.

Topics such as *Uniform Civil Code (UCC)* and *Brain Drain* show a relatively higher proportion of supportive tweets (stance 1), suggesting favorable public perception or concern acknowledgment. Conversely, topics like *Animal Sacrifices* and *Citizenship Amendment Act (CAA)* demonstrate more polarized or neutral distributions, indicating contentious or sensitive public discussions.

LGBTQ and *Anti-Conversion Laws* exhibit relatively balanced stances across the spectrum, implying ongoing debate and diverse opinions in the Indian Twitter space.

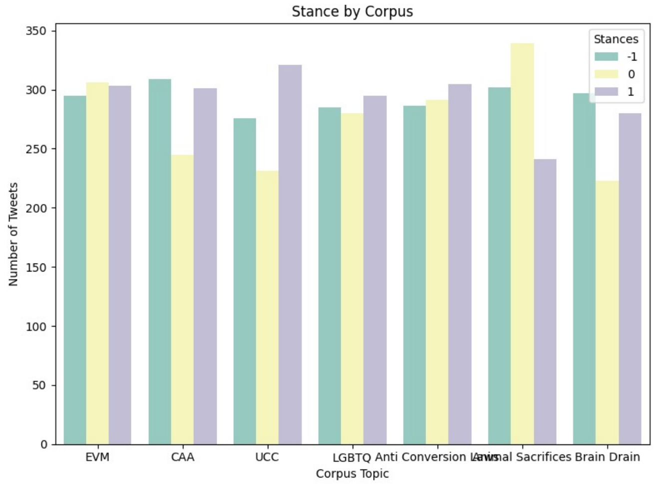

Figure 1. Stance Distribution by Corpus Topic. Each bar cluster represents a topic from the dataset, with the three bars indicating the number of tweets labeled as opposing (-1), neutral (0), and supporting (1).

Electronic Voting Machines (EVM) also reveals a near-equal distribution, highlighting persistent uncertainty or skepticism among users.

Overall, this stance-wise breakdown reflects the moral and political complexity embedded in digital public discourse in India, which has implications for both stance detection and moral sentiment analysis tasks.

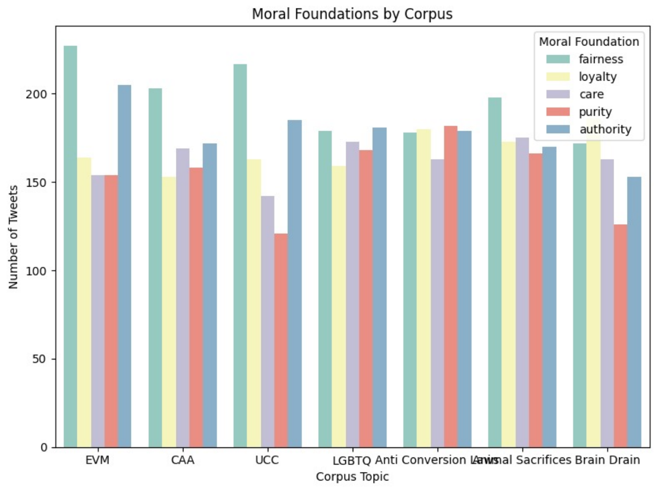

Figure 2. Moral Foundations by Corpus Topic. Each bar cluster represents a topic, and the bars within each cluster indicate the number of tweets invoking a specific moral foundation: fairness, loyalty, care, purity, and authority.

Figure 2 illustrates the distribution of tweets annotated with moral foundations across seven distinct socio-political topics within the Indian Twitter corpus. The five moral foundations considered in this analysis are **Fairness**, **Loyalty**, **Care**, **Purity**, and **Authority**, as defined by Moral Foundations Theory.

Across the topics, **Fairness** consistently emerges as the most prevalent foundation, particularly dominant in tweets discussing *EVM* and *UCC*, reflecting concerns about procedural justice, equity, and inclusivity. Tweets related to *LGBTQ* and *Anti-Conversion Laws* show a relatively balanced invocation of *care*, *purity*, and *fairness*, suggesting that discussions on these issues are shaped by multifaceted moral concerns.

The topic of *Animal Sacrifices* sees heightened references to *purity*, a foundation often associated with traditions and sanctity norms, while *Brain Drain* elicits strong responses grounded in *authority* and *fairness*, pointing to critiques of systemic governance or institutional integrity.

This analysis underscores the utility of moral foundation tagging in highlighting the underlying value-based reasoning within digital discourse. The results suggest that different socio-political topics in the Indian context evoke distinct configurations of moral reasoning, which can be leveraged to enhance models of moral sentiment classification.

To ensure the accuracy and consistency of annotation, label assignments—both sentiment and moral foundation—were verified by two independent human annotators. Discrepancies were resolved through mutual discussion. In addition, language models similar to ChatGPT were employed to assist in the disambiguation of complex or ambiguous tweets, providing additional validation and contextual support. This multi-layered annotation approach combines human expertise with LLM capabilities to strengthen the reliability of the dataset and enhance the depth of subsequent model training and analysis. This distribution of topics reflects the key social, political, and cultural issues that are currently being discussed within the Indian context. It provides insights into the concerns that shape public debates and policy discussions, offering a snapshot of the themes that are most salient in Indian society today.

Link: https://github.com/nlvmadhav/Indian-Tweet-Data

Methodology

To accommodate the dual tasks of sentiment classification and moral foundation identification, we designed two parallel pipelines with rigorous fine-tuning and parameter search strategies.

Text Preprocessing

Raw tweets undergo a multi-stage cleaning process: lowercasing, removal of URLs, user mentions, emojis, HTML tags, non-ASCII characters, and extra whitespace. Hashtags and contractions are expanded, and stopwords are filtered. For the Bi-LSTM pipeline, tokenization is performed using the NLTK tokenizer, while for BERT, we leverage the WordPiece tokenizer with a maximum sequence length of 128 tokens.

Method 1: TF-IDF + Bi-LSTM Text Classification Pipeline

To perform tweet classification, we implemented a hybrid pipeline that combines traditional feature extraction (TF-IDF) with deep learning (Bi-LSTM) to leverage both statistical and sequential aspects of textual data. This section details the methodology, including hyperparameter tuning and architectural choices.

1. Text Vectorization using TF-IDF: The first stage of the pipeline involves converting raw text into numerical feature vectors using the **Term Frequency-Inverse Document Frequency (TF-IDF)** method. This approach assigns weights to terms based on their frequency within a document relative to their frequency across the entire corpus, effectively highlighting terms that are more informative for a specific document. To ensure a robust and computationally feasible representation, we tuned several hyper-parameters:

- **Max Features:** We experimented with maximum feature limits of *5,000*, *10,000*, and *20,000*. This parameter controls the size of the vocabulary used for vectorization. Lower values help reduce dimensionality and memory requirements, while higher values preserve more expressive detail.
- **Feature Selection:** After vectorization, we applied feature selection techniques to reduce noise and improve generalization. Specifically, we used *Chi-square statistics* and *mutual information* to rank features by their discriminative power with respect to class labels. We retained the *top 80%* of ranked features to mitigate overfitting and enhance training efficiency.

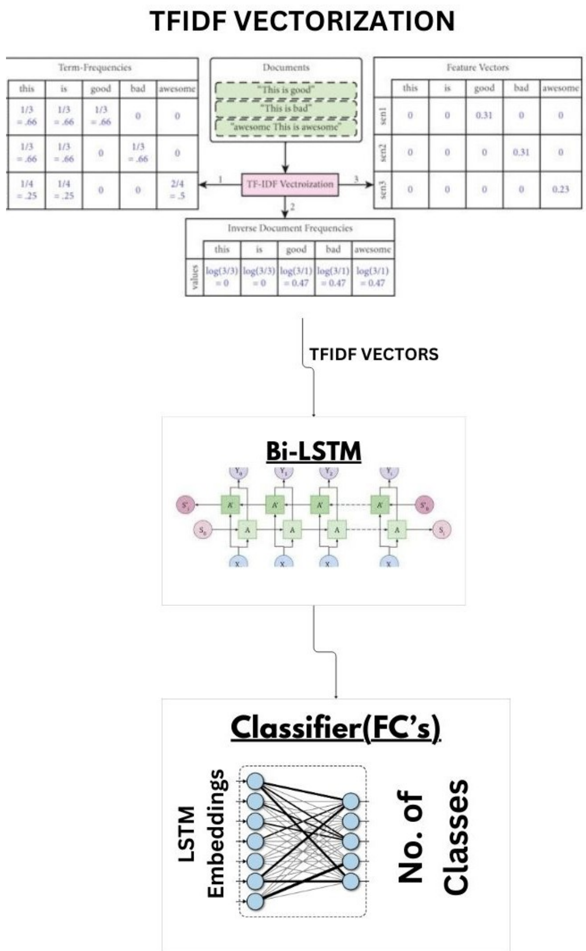

Figure 3. TF-IDF + Bi-LSTM Pipeline

2. Bi-LSTM Neural Network Architecture: The refined TF-IDF vectors were used as inputs to a **Bidirectional Long**

Short-Term Memory (Bi-LSTM) network, which is well-suited for learning long-term dependencies and capturing the contextual flow of text.

- **Bi-LSTM Layers:** The model consists of two stacked Bi-LSTM layers with a *hidden size of 256*. The bidirectional setup allows the model to consider both past and future context, enhancing understanding of sentence structure.
- **Dense Layers:** The Bi-LSTM output is passed through two fully connected *dense layers* with *128* and *64 units*, respectively. Both layers use the *ReLU activation function* to introduce non-linearity, enabling the model to learn more complex patterns.
- **Dropout Regularization:** A *dropout rate of 0.3* is applied after each layer to prevent overfitting. This technique randomly deactivates a portion of neurons during training to improve generalization.

3. Training Configuration:

- **Optimizer and Learning Rate:** We used the *Adam optimizer*, known for its adaptive learning capabilities. The learning rate was tuned in the range of *1e-3 to 5e-4* based on validation performance.
- **Early Stopping:** To avoid overfitting and unnecessary training, we implemented *early stopping* based on the *validation F1-score*. Training was halted if no improvement was observed for *3 consecutive epochs*, with a maximum of *20 epochs*.
- **Batch Size:** We tested *batch sizes of 32 and 64*, selecting the best configuration based on training speed and memory utilization, especially for optimal *GPU efficiency*.

This pipeline effectively combines the interpretability and simplicity of TF-IDF with the powerful sequence modeling capabilities of Bi-LSTM. It is especially suitable for short and context-sensitive text like tweets. Through careful tuning and regularization, the model aims to achieve strong generalization across a range of nuanced and diverse topics.

Method 2:Bidirectional Gated Recurrent Unit (Bi-GRU)

The sentiment and morality classification task was also approached using a **Bidirectional Gated Recurrent Unit (Bi-GRU)** model, which is effective in modeling sequential data such as text. Bi-GRUs are a simplified variant of Bi-LSTMs and are capable of capturing long-term dependencies with fewer parameters and reduced computational overhead.

The implemented model, `Bi-GRUClassifier`, accepts input vectors of dimension 300, corresponding to the dimensionality of the pre-trained GloVe embeddings. The model includes the following components:

- A **bidirectional Bi-GRU layer** with a hidden size of 128 units in each direction. The bidirectionality enables the model to learn from both past and future contexts of the input sequences, improving its ability to understand sentiment and moral cues in tweets.
- A **fully connected linear layer** that maps the concatenated hidden states from the forward and backward Bi-GRU layers (total 256 dimensions) to the output dimension (three sentiment classes: for, against, neutral).
- A **dropout layer** to reduce the likelihood of overfitting by randomly deactivating neurons during training.

The forward pass involves encoding the tweet as a sequence of GloVe embeddings, processing it through the Bi-GRU layer, concatenating the last hidden states from both directions, applying dropout, and finally generating class scores via the linear layer.

GloVe Embedding Integration

The input to the Bi-GRU model consists of word embeddings loaded from pre-trained **GloVe vectors**. These embeddings were specifically trained on Twitter data to capture the informal and concise nature of social media language. The loading function reads the GloVe file and stores each word and its corresponding embedding vector in a dictionary structure.

The loading function operates as follows:

- It reads the embedding file line by line.
- Each line is split into the word and its embedding values.
- The word serves as the key, and its vector (converted to a NumPy array) is stored as the value.

This embedding dictionary is then used to initialize the embedding layer or to map tokens from tweets to their respective GloVe vectors during preprocessing. The use of GloVe embeddings provides semantically meaningful representations that help the Bi-GRU model understand the underlying sentiment and morality expressed in the tweet texts.

Method 3: Transformer-Based Classification: BERT Fine-Tuning

To further improve classification performance on our tweet dataset, we employed a transformer-based approach by fine-tuning the pre-trained **BERT-base-uncased** model. BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. (2019), is a deep contextualized language model that captures bidirectional dependencies and has demonstrated state-of-the-art performance across various NLP tasks. The BERT-base model contains 12 transformer encoder layers, 768 hidden dimensions, and approximately 110 million parameters. Fine-tuning pre-trained transformers involves adapting the generic language representations to the target task with minimal supervised training data. To achieve optimal performance and stability during training, we adopted several widely accepted fine-tuning strategies:

1. Batch Size and Gradient Accumulation: Given GPU memory constraints, we used batch sizes of 16 and 32, combined with **gradient accumulation over 2 steps** to simulate larger effective batch sizes. Gradient accumulation allows gradients to be accumulated over multiple mini-batches before performing a weight update, which is especially beneficial when training deep models on limited hardware. This technique maintains training stability and

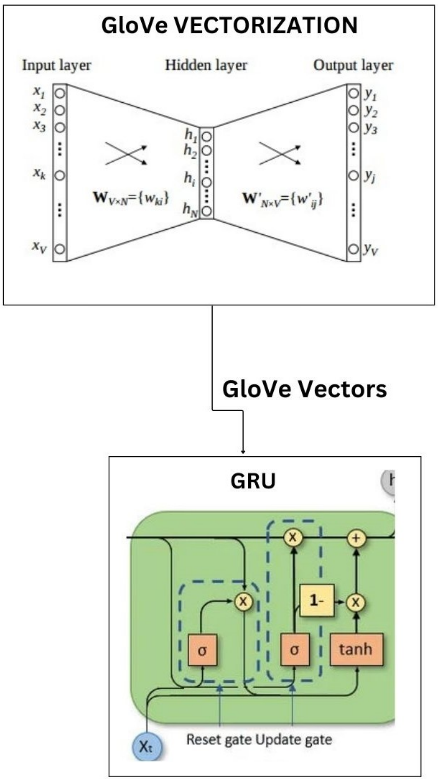

Figure 4. Bi-GRU + GloVe Encoding

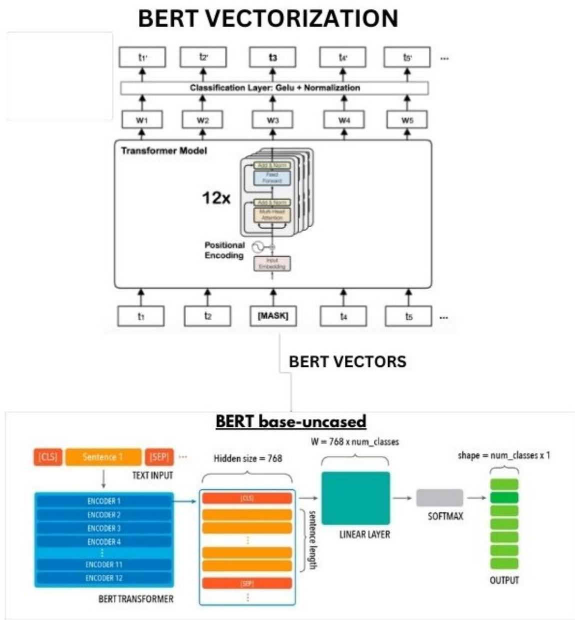

Figure 5. BERT Architecture

provides the generalization benefits of larger batch sizes without exceeding memory limitations.

2. Regularization Techniques: To prevent overfitting, we applied multiple forms of regularization:

- **Weight Decay:** A coefficient of 0.01 was applied to all trainable parameters (excluding biases and normalization layers) to encourage smaller, more generalizable weights.
- **Dropout:** A dropout rate of 0.1 was applied in the final classification head to reduce overfitting by randomly deactivating neurons during training.

These methods help the model maintain robustness, especially when trained on short, informal text such as tweets.

3. Optimization Strategy: The model was optimized using the **AdamW** optimizer, which decouples weight decay from the gradient update rule. This optimizer is standard for transformer fine-tuning and is known to improve training stability. We used:

- $\epsilon = 1 \times 10^{-8}$ for numerical stability,
- **Gradient clipping** with a maximum norm of 1.0 to prevent exploding gradients.

4. Training Duration and Early Stopping: Fine-tuning was conducted for **3 to 5 epochs**, which is generally sufficient for convergence in low-resource scenarios. We used **early stopping** based on validation loss, terminating training when no improvement was observed for two consecutive epochs. This prevents overfitting and ensures efficient use of computational resources. This fine-tuning protocol combines proven techniques in transformer-based NLP with task-specific adaptations for efficient training and robust generalization. The use of progressive unfreezing, learning rate scheduling, gradient accumulation, and regularization collectively enables effective adaptation of BERT to the nuances of social media text classification.

Method 4: Gated Recurrent Unit (Bi-GRU) with TF-IDF Embeddings

This approach utilizes a **Bidirectional Gated Recurrent Unit (Bi-GRU)** model combined with **TF-IDF embeddings** to evaluate the effectiveness of traditional text representation techniques in sentiment and moral foundation classification tasks.

Unlike dense embeddings such as GloVe, **TF-IDF (Term Frequency-Inverse Document Frequency)** represents each tweet as a sparse vector that highlights the importance of words based on their frequency within the dataset. This method preserves important features while reducing noise from commonly occurring words.

The implemented model, `Bi-GRU_TFIDF_Classifier`, consists of the following key components:

- A **Bi-GRU layer** with a hidden size of 128 units. While TF-IDF vectors do not inherently capture word order, the Bi-GRU processes the vectorized input to learn relevant patterns across features.
- A **fully connected linear layer** that maps the Bi-GRU's hidden state to the output classes—three sentiment labels (favor, against, neutral) or five moral foundations (care, fairness, loyalty, authority, purity), depending on the classification task.
- A **dropout layer** to prevent overfitting by randomly deactivating neurons during training.

During the forward pass, each tweet is first converted into a TF-IDF feature vector. This vector is then fed into the Bi-GRU layer, which processes the input to capture meaningful relationships between features. The resulting hidden state is passed through the dropout layer before being fed into the linear layer to generate the final class predictions.

This method allows for a direct comparison between traditional TF-IDF representations and modern embedding techniques, providing insights into how simpler, interpretable features perform when combined with recurrent neural networks for sentiment and morality classification tasks.

Experimental Analysis and Results

Sentiment Analysis Performance

To evaluate the effect of different neural architectures and embedding strategies on sentiment classification, we conducted a comparative analysis involving five model configurations: BERT, Bi-LSTM with TF-IDF embeddings, Bi-GRU with GloVe embeddings, Bi-GRU with TF-IDF embeddings, Bi-LSTM with GloVe embeddings. The classification task was to label tweets into one of mentioned three sentiment categories—*favor*, *against*, or *neutral*. The results are presented in Table 2.

Table 2. Sentiment Classification Results

Model	Embeddings	Accuracy(%)	Precision	Recall	F1-score
BERT	BERT	99.19	0.9919	0.9919	0.9918
Bi-LSTM	TF-IDF	78.24	0.7836	0.7824	0.7825
Bi-GRU	GloVe	75.64	0.7694	0.7564	0.7824
Bi-GRU	TF-IDF	70.01	0.7079	0.7001	0.7002
Bi-LSTM	GloVe	32.61	0.1064	0.3261	0.1604

From the results in Table 2, it is evident that the transformer-based BERT model outperforms all other architectures across all metrics, with a near-perfect accuracy of 99.19%. This substantial performance is attributed to BERT's contextualized embeddings and bidirectional attention mechanisms, which enable deep understanding of sentiment context.

The Bi-LSTM model with TF-IDF embeddings achieves the second-highest accuracy at 78.24%, indicating that bidirectional memory in RNNs is effective, although its reliance on sparse embeddings limits semantic representation.

The Bi-GRU with GloVe embeddings achieves an accuracy of 75.64%, illustrating that Bi-GRUs can approximate Bi-LSTM performance when equipped with dense, pre-trained embeddings.

The Bi-GRU with TF-IDF embeddings achieves an accuracy of 70.01%, illustrating that sparse, high-dimensional vectors limit performance, even in capable architectures like Bi-GRUs.

Finally, the implementation of Bi-LSTM with GloVe embeddings proves with the least readings of all the working models with accuracy of 32.61% in sentiment analysis classification.

Moral Foundation Classification Performance

We also evaluated the models for moral foundation classification using the same four configurations. The task was to classify each tweet into one of five moral dimensions—Care, Fairness, Loyalty, Authority, and Purity. The comparative results are provided in Table 3.

Table 3. Moral Foundation Classification Results

Model	Embeddings	Accuracy(%)	Precision	Recall	F1-score
BERT	BERT	91.73	0.9182	0.9173	0.9175
Bi-LSTM	TF-IDF	62.44	0.6275	0.6244	0.6250
Bi-GRU	GloVe	53.15	0.5614	0.5315	0.5315
Bi-GRU	TF-IDF	45.73	0.4733	0.4573	0.4516
Bi-LSTM	GloVe	28.34	0.1023	0.2834	0.1156

From Table 3, the BERT model again shows superior performance with an accuracy of 91.73%, followed by Bi-LSTM with TF-iDF with an accuracy of 62.44%. The Bi-GRU model with GloVe embeddings yields 53.15%, and the Bi-GRU with TF-IDF performs with about a low accuracy of 45.73% and finally, Bi-LSTM with GloVe embeddings again performs the lowest at 28.34%.

Similar to sentiment classification, models with TF-IDF embeddings outperform those with GloVe, highlighting the effectiveness of high-dimensional representations in tasks involving subjective interpretation with local context data. However, due to the nuanced and culturally specific nature of moral language, simpler architectures like Bi-GRU struggle with generalization, even when supported by pre-trained embeddings.

In conclusion, the study highlights the critical role of both model architecture and embedding type in effectively processing complex, localized datasets like Indian Twitter content. The transformer-based BERT model consistently outperformed all other models in sentiment and moral foundation classification, owing to its deep contextual understanding and bidirectional attention mechanism. Among recurrent neural networks, Bi-LSTM paired with TF-IDF embeddings delivered better results, followed by Bi-GRU with GloVe embeddings, Bi-GRU with TF-IDF embeddings, and Bi-LSTM with GloVe embeddings.

Conclusion and Future Work

Despite the inherent challenges posed by the informal, colloquial, nature of Twitter discourse, the proposed models exhibited strong performance. These findings underscore the efficacy of state-of-the-art natural language processing (NLP) techniques in capturing nuanced moral and attitudinal signals within short social media texts. The framework makes a notable contribution to computational social science, especially in multicultural and politically dynamic settings such as India.

Our experimental evaluation underscores the better performance of the transformer-based BERT model, compared to all other architectures with accuracy values of 99.19% for sentiment analysis and 91.73% for detecting moral foundations. These accuracies can be explained due to BERT's capacity for utilizing deep contextual embeddings as well as fine-tuning, making it highly appropriate for analyzing the subtle, culturally saturated language of Indian socio-political discourse.

Among the recurrent models, Bi-LSTM with TF-IDF embeddings demonstrates moderate but reliable performance, achieving 78.24% accuracy for sentiment and 62.44% for morality. The Bi-LSTM model benefits from its capacity to capture bidirectional dependencies, offering a practical trade-off between interpretability, resource efficiency, and predictive accuracy—especially valuable in low-resource or deployment-constrained environments.

The Bi-GRU model with GloVe embeddings is somewhat lower (75.64% for sentiment and 53.15% for morality) but still an attractive lightweight option. Its use of pretrained dense word vectors and streamlined architecture provides a trade-off between performance and computational expense, allowing it to be feasible for edge deployment or environments without the resources for transformer models.

Conversely, the Bi-GRU model based on TF-IDF embedding is always at the bottom of the ranks with the lowest scores of 70.01% in sentiment and 45.73% in morality classification. This inferiority highlights the inadequacies of sparse feature representation and the unidirectional Bi-GRU's shallow modeling depth when it comes to capturing abstract or context-dependent moral reasoning. Finally, the implementation of Bi-LSTM with GloVe embeddings proves with the least readings of all the working models with accuracy of 32.61% and 28.34% respectively.

Significantly, this paper adds not merely model findings, but also an enriching dataset to be drawn upon in later work. Our hand-curated Indian Twitter-based corpus allows future work to build upon moral sentiment in Indian Twitter conversations and serves to train models or supplement modest, domain-aligned annotation efforts directly. Because moral sentiment is so culturally relative and context-dependent, particularly in extremely polarized political times, the mixture of general-purpose annotations with targeted data will likely prevent sparsity and enhance model generalizability.

Ultimately, this project highlights the strength of transformer-based models in subtle sentiment and moral grounding analysis, yet acknowledging the practical significance of recurrent architecture in actual low-resource settings. It paves the way towards building scalable real-time tools to track and understand public debate, providing meaningful insights into the moral and affective currents governing Indian social media debates which does not restrict to a generalized global context issues and expand to any local contextual problems(as India in this case).

While the current framework demonstrates promising results, several potential areas for improvement and extension remain:

- **Increased Dataset Diversity:** Expanding the dataset to include a broader range of topics and tweets in additional Indian languages will enhance the model's generalizability across linguistic and cultural boundaries.
- **Detection of Implicit and Sarcastic Expressions:** Future work should focus on improving the system's capacity to identify and interpret implicit, ambiguous, or sarcastic stances, which are often difficult to classify using standard models.
- **Platform-Specific Signal Integration:** Incorporating Twitter-specific features—such as hashtags, mentions, emojis, and retweet patterns—could provide supplementary context that improves classification accuracy.
- **Real-Time Processing Capability:** Enhancing the framework to support real-time data analysis would facilitate applications such as live sentiment monitoring, social media moderation, and trend detection on high-volume streams.
- **API Development and Integration:** Building a stable, developer-friendly API will enable integration into external platforms and tools, allowing practical use cases in misinformation detection, hate speech identification, and public opinion tracking.

Future work will emphasize boosting the framework's accuracy, scalability, and adaptability across multiple languages and platforms. The overarching goal is to support large-scale, real-time, and ethically-grounded evaluation of moral sentiment in digital discourse, contributing to healthier online interactions and more informed public dialogue.

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